

Thermo–Richards–Mechanics (TRM) Open-Geo-Sys model

MB

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1 Link to paper

This is summary of the configured OGS TRM model, rewritten into Wang–Kosakowski–Kolditz (2009) notation.

Discussion goals:

- (i) confirm which couplings we actually switch on in OGS, and
- (ii) see clearly how each XML parameter lands in the PDEs.

2 Topics and issues to discuss in Leipzig (or earlier)

The plan: based on provided data, present a model tuned for Bayesian inversion to determine $\mathbf{k}$. The current audit resolves the two formulation comments that were still open in the first version of the notes:

- **Homogeneity versus heterogeneity:** the exchanged OGS model remains a homogeneous-material TRM model. The active workflow introduces heterogeneity only through run-local mesh-cell parameter fields, first for the intrinsic permeability tensor $\mathbf{k}(\mathbf{x})$. Porosity and retention parameters are documented below as later releasable scalar options, but they are not part of the active objective until OGS state outputs and identifiability checks exist.
- **Specific expression for $k_{\text{rel}}(S)$:** the XML uses `RelativePermeabilityVanGenuchten` with $S_{lr} = 0.1$, $S_{gr} = 0$, $m = 0.45$, and $k_{\text{rel}}^{\min} = 10^{-25}$. The explicit function is given in Table 3; it is fixed in the current permeability-focused workflow.

The remaining unresolved model-provenance items are not equation-formulation issues: the suspicious active CTE value, the provenance of the open-niche RH pressure curve, the native-host OGS executable caveat (the collected OGS 6.5.4 SIF is used through the recorded Dockerized-Apptainer backend), and the still-pending ERT and Taupe/TDR observation-operator calibrations are tracked in the measurement and workflow sections.

3 Model provenance and non-modification constraint

The editable report folder contains the LaTeX source and a copy of the OGS settings. For the model audit, the authoritative exchanged inputs are the Gesa TeamBeam archives copied into `GESA_model_original/2025-04-03_CDA_N4_2D_250403` and `GESA_model_original/2025-05-09_CDA_N4_2D_250509`. The report’s `ogs_settings` directory matches the May 2025 XML settings, with the project file `cd_a.open.niche.quad.prj` retained from the April archive. The April archive also contains the mesh files (`bulk.vtu`, `bulk_all.vtu`, and the boundary meshes), while the May archive updates the timestep/output schedule to align outputs with monthly ERT data.

The model itself is treated here as frozen. Measurement information should therefore enter through (i) observation operators applied to the model outputs, (ii) preprocessing of boundary-condition curves or comparison targets, and (iii) externally generated parameter-field files or XML parameter replacements used by an inversion workflow. Those generated fields can vary between inversion samples, but the mathematical meaning of the OGS process and the original mesh/project structure should remain explicit.

This non-modification claim is also checked by the generated formulation audit `inversion_workflow/ogs_formulation_consistency_audit.md`. That audit parses the source XML and the representative run-local XML, verifies 18 formulation checks with no hard failures, and records the two remaining caveats separately: `phi` is a run-local mesh field but fixed at 0.105, and the active CTE value still needs provenance confirmation. The run-local workflow therefore substitutes `k_i_r_d` for the heterogeneous intrinsic-permeability tensor without changing the TRM process, phase laws, media laws, or boundary-condition semantics.

4 Governing equations (Wang 2009 notation)

Primary variables: displacement $\mathbf{u}$, liquid pressure p , temperature T . Porous medium: porosity n , saturation S , intrinsic permeability $\mathbf{k}$, relative permeability $k_{\text{rel}}(S)$, liquid density ρ_w , viscosity μ .

Darcy velocity and mass flux (unsaturated, gravity off): The OGS TRM process uses the Darcy velocity

$$\mathbf{v}^l = -\frac{k_{\text{rel}}(S)\mathbf{k}}{\mu}\nabla p, \quad \mathbf{q}_w = \rho_w \mathbf{v}^l,$$

because `specific_body_force` is zero in `01_processes_TRM.xml`. The factors nS belong to storage and thermal mixing terms, not to the Darcy velocity itself [15, Sec. “Governing equations without vapor diffusion”].

Liquid mass balance (no vapor): For the actual OGS TRM formulation without vapor diffusion, the hydraulic equation is

$$\left(S\beta - n\frac{\partial S}{\partial p_c}\right)\rho_w \dot{p} - S\left(\frac{\partial \rho_w}{\partial T} + \rho_w(\alpha_B - S)\alpha_T^s\right)\dot{T} + \nabla \cdot (\rho_w \mathbf{v}^l) + S\alpha_B \rho_w \nabla \cdot \dot{\mathbf{u}} = Q_H, \quad p_c = -p.$$

This resolves the earlier comment on β_p : the XML does not use constant density. The liquid density is the OGS **Linear** model

$$\rho_w(T, p) = 1095 \left[1 - 5 \times 10^{-4}(T - 298.15) + 3.2 \times 10^{-10}(p - 7.5 \times 10^6)\right],$$

so $\partial \rho_w / \partial p \neq 0$ and $\partial \rho_w / \partial T \neq 0$. The OGS storage term is not the informal $S\rho_w^{-1}\partial \rho_w / \partial p$ expression alone; it is the composite $S\beta$ term combined with the capillary-storage term $-n\partial S / \partial p_c$. Likewise, vapor diffusion is absent, but the no-vapor OGS TRM equation still contains a thermal coupling term through liquid density temperature dependence and solid thermal expansivity [25, pp. 1633–1634] [15, Sec. “Governing equations without vapor diffusion”].

Momentum balance (quasi-static):

$$\nabla \cdot (\boldsymbol{\sigma} - b(S)\alpha_B p \mathbf{I}) + \mathbf{f} = \mathbf{0}, \quad \boldsymbol{\sigma} = \mathbf{C} : \left(\boldsymbol{\varepsilon}(\mathbf{u}) - \boldsymbol{\varepsilon}^T - \boldsymbol{\varepsilon}^{\text{sw}}(S)\right).$$

In this model $\mathbf{f} = \mathbf{0}$, $\alpha_B = 1$, and `BishopsPowerLaw` with exponent 1 gives $b(S) = S$. The compact form $\nabla \cdot (\boldsymbol{\sigma} - Sp\mathbf{I}) = \mathbf{0}$ is therefore valid only after those XML substitutions. Wang et al. write the thermal stress term directly for an isotropic special case; OGS instead places thermal strain inside the generalized Hooke law, which is the correct representation for the configured orthotropic `LinearElasticOrthotropic` relation and the active saturation-dependent swelling term [25, pp. 1633–1634] [15, Sec. “Governing equations without vapor diffusion”].

Heat balance:

$$(\rho c_p)^{\text{eff}} \dot{T} - \nabla \cdot (\mathbf{k}_T^{\text{eff}} \nabla T) + \rho_w c_w \nabla T \cdot \mathbf{v}^l = Q_T,$$

with OGS effective phase mixing

$$(\rho c_p)^{\text{eff}} = (1 - n)\rho_s c_s + nS\rho_w c_w, \quad \mathbf{k}_T^{\text{eff}} = (1 - n)\mathbf{k}_T^s + nS\mathbf{k}_T^l \mathbf{I}.$$

This resolves the heat-balance comment: the missing nS is in storage and thermal conductivity mixing, while OGS writes the advective heat term with the Darcy velocity $\mathbf{v}^l$, not with pore velocity. Expanding $\nabla \cdot (\rho_w c_w \mathbf{v}^l T)$ into $\rho_w c_w \mathbf{v}^l \cdot \nabla T$ is only valid after using the mass balance and the OGS weak-form assumptions; for the report we keep the official OGS form rather than an independently expanded flux form [15, Sec. “Governing equations without vapor diffusion”].

5 What differs from Wang et al. (2009)

- **Coupling/solver:** one monolithic `THERMO_RICHARDS_MECHANICS` process with Backward Euler time discretization and one Newton solver.
- **Gravity:** disabled here ($\mathbf{g} = \mathbf{0}$); no hydrostatics.
- **Vapor:** neglected here; Wang derives/keeps vapor terms.
- **Elasticity:** orthotropic linear elasticity here vs. isotropic demos in Wang.
- **Bishop/Biot:** $b(S) = S$ and $\alpha = 1 \Rightarrow$ pore-stress term $Sp\mathbf{I}$ (same structure).
- **Swelling:** saturation-dependent swelling strain active here.
- **Thermal conductivity:** OGS effective porosity mixing for k (solid+liquid; no gas).

6 Boundary data

- **Temperature:** Dirichlet $T = 298.15$ K on `bulk_all`.
- **Pressure:** Dirichlet $p = 1.5$ MPa on `cd-a_top`; time-dependent `open_niche_seasonal` pressure on `cd-a_niche4`. The parameter `bc_pressure_outside` is defined but is not attached to an active boundary condition.
- **Mechanics:** zero normal displacement on the outer box (`left/right` in component 0, `top/bottom` in component 1). `load_top` and `initial_stress` are defined/commented but inactive.

7 XML $\rightarrow$ PDE symbol mapping

Below, each XML entry is mapped to the PDEs parameter.

Table 1: Medium/phases: properties and laws mapped to PDE symbols

XML path (name)	Type	Symbol	Unit	Where it appears / comment
porosity (phi)	Parameter	n	–	Storage in mass balance; mixing in thermal storage/conductivity.
permeability (k_i)	Param (tensor)	$\mathbf{k}$	m^2	Darcy flux; anisotropy allowed.
biot_coefficient (biot)	Parameter	α	–	Pore-stress partition (with Bishop $b(S) = S$).
thermal_conductivity (mixing)	Model	k	$\text{W m}^{-1}\text{K}^{-1}$	Conductive heat flux; porosity mix of solid+liquid.
viscosity (eta)	Parameter	μ	Pa s	Flow resistance in Darcy’s law.
density (Linear)	Model	$\rho_w(T, p)$	kg m^{-3}	Storage/advection terms; linearized about a reference state.
specific_heat_capacity (c_p_l)	Parameter	c_w	$\text{J kg}^{-1}\text{K}^{-1}$	Enters $(\rho c)_r$ and heat advection.
thermal_conductivity (kappa_l)	Parameter	k_ℓ	$\text{W m}^{-1}\text{K}^{-1}$	Input to the effective k mixing.
density (rho_s)	Parameter	ρ_s	kg m^{-3}	Thermal storage mixing (solid fraction).
specific_heat_capacity (c_p_s)	Parameter	c_s	$\text{J kg}^{-1}\text{K}^{-1}$	Thermal storage mixing (solid fraction).
thermal_conductivity (kappa_th_s)	Parameter	k_s	$\text{W m}^{-1}\text{K}^{-1}$	Input to effective k .
thermal_expansivity (CTE)	Parameter	α_T	K^{-1}	Thermal strain and hydraulic thermal-storage coupling; the active XML value is suspicious and must not be interpreted as calibrated.
swelling_stress_rate	Model	$\epsilon^{\text{sw}}(S)$	–	Saturation-dependent swelling strain (isotropic; limits in XML).
relative_permeability (VG)	Model	k_{rel}	–	Hydraulic conductivity factor in Darcy’s law.
saturation (VG)	Model	$S(p_c)$	–	Storage term via $\partial S/\partial p$.
bishops_effective_stress (power=1)	Model	$b(S) = S$	–	Pore-stress scaling in momentum balance.

Table 2: ICs/BCs: parameters mapped to variables and equations

XML path (name)	Type	Applies to	Unit	Meaning / where used
ic_displacement	Constant	$\mathbf{u}(\cdot, 0)$	m	Initial displacement (typically $\mathbf{0}$).

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Table 2: ICs/BCs: parameters mapped to variables and equations (continued)

XML path (name)	Type	Applies to	Unit	Meaning / where used
ic_pressure	Function	$p(\cdot, 0)$	Pa	Initial liquid pressure (current function yields a constant).
ic_temperature	Constant	$T(\cdot, 0)$	K	Initial temperature.
bc_displacement	Constant	$\mathbf{u}$	m	Zero-displacement box (kinematic constraints).
bc_top_pressure	Function	p	Pa	Dirichlet pressure on top boundary.
open_niche_seasonal	CurveScaled	p	Pa	Time-varying pressure on niche boundary.
bc_temperature_outside	Constant	T	K	Outer temperature Dirichlet.
load_top	Function	traction	Pa	Defined but inactive in the process file.
bc_atm_pressure	Constant	ref.	Pa	Reference value; not bound to an active BC.

8 Model parameters, fixed values and candidate priors

In this section we collect all material parameters entering the TRM model and split them into: (i) parameters and constitutive laws that are kept fixed from the exchanged model for the present workflow, and (ii) parameters that could be released in a later Bayesian calibration once the relevant observation operators are active. The only implemented field unknown at present is the intrinsic permeability tensor $\mathbf{k}(\mathbf{x})$, treated through run-local mesh-cell data. The sampled NMR state residual is now part of the active objective, but porosity remains a fixed support field until the NMR residual policy and porosity–saturation separation gate are approved. Retention and mechanical parameters are listed as later candidate priors, not as already-fitted active unknowns. The machine-readable staging decision is regenerated by `inversion_workflow/scripts/build_inversion_parameter_release_plan.py` and stored in `inversion_workflow/inversion_parameter_release_plan.md`.

Table 3: Fixed material and fluid parameters and constitutive laws (values taken from the XML configuration).

XML name / property	Symbol	Unit	Value / definition
<i>Solid phase (Opalinus Clay matrix)</i>			
rho_s (density)	ρ_s	kg m^{-3}	Dry solid density, taken constant: $\rho_s = 2.52 \times 10^3$.
kappa_th_s (thermal_conductivity)	k_s	$\text{W m}^{-1}\text{K}^{-1}$	Solid thermal conductivity: $k_s = 2.156$.
c_p_s (specific_heat_capacity)	c_s	$\text{J kg}^{-1}\text{K}^{-1}$	Solid specific heat capacity: $c_s = 1254.74$.
CTE (thermal_expansivity)	α_T	K^{-1}	Active XML value $\alpha_T = 1.25474 \times 10^3$, with the XML comment incorrectly labelling it as a heat-capacity-like unit. This is not physically plausible for a rock thermal expansion coefficient and is likely a copied placeholder from <code>c_p_s</code> . The generated thermal-expansivity audit compares this value with HE-D/Opalinus Clay reference scales: Garitte et al. list linear solid thermal-expansion coefficients from 1.0×10^{-5} to $2.6 \times 10^{-5} \text{ K}^{-1}$, and Gens et al. used $1.4 \times 10^{-5} \text{ K}^{-1}$ in thermal-loading interpretation [6, PDF p. 12][9, PDF p. 14]. The XML value is therefore 4.83×10^7 times the high end of that range. Because the current run is nearly isothermal, the hydraulic calibration can proceed only if this parameter is treated as an active-model caveat, not as a calibrated Opalinus Clay property.

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Table 3: Fixed material and fluid parameters and constitutive laws (continued).

XML name / property	Symbol	Unit	Value / definition
swelling_stress_rate (SaturationDepen- dentSwelling)	$\varepsilon^{\text{sw}}(S)$	–	Isotropic saturation-dependent swelling strain defined by the OGS SaturationDependentSwelling model with swelling pressures $p_{\text{sw}} = 1.0 \times 10^6$ Pa in all principal directions, exponents 1, 1, 1, and activation for $S \in [0.1, 1]$. The detailed functional form is kept fixed as in the XML.
E, G, nu (orthotropic elasticity)	$\mathbf{C}$	Pa, –	Orthotropic linear elasticity as in LinearElasticOrthotropic : Young moduli $E_1 = 13.8 \times 10^9$ Pa, $E_2 = 6.0 \times 10^9$ Pa, $E_3 = 13.8 \times 10^9$ Pa; shear moduli $G_1 = 4.79 \times 10^9$ Pa, $G_2 = 2.46 \times 10^9$ Pa, $G_3 = 4.79 \times 10^9$ Pa; Poisson ratios $\nu_{12} = 0.44$, $\nu_{23} = 0.22$, $\nu_{13} = 0.44$. These are kept fixed in the hydraulic-focused inversion.
<i>Liquid phase (aqueous pore water)</i>			
c_p_l (specific_heat_capacity)	c_w	J kg ^{−1} K ^{−1}	Liquid specific heat capacity: $c_w = 4160$.
kappa_l (thermal_conductivity)	k_ℓ	W m ^{−1} K ^{−1}	Liquid thermal conductivity: $k_\ell = 0.6$.
eta (viscosity)	μ_w	Pa s	Dynamic viscosity of the liquid: $\mu_w = 1.0 \times 10^{-3}$, constant (temperature dependence neglected in this configuration).
density (Linear)	$\rho_w(T, p)$	kg m ^{−3}	Liquid density given by an affine law in temperature and pressure: $\rho_w(T, p) = \rho_{w,0} \left(1 + a_T (T - T_0) + a_p (p - p_0) \right),$ with reference value $\rho_{w,0} = 1095$, $T_0 = 298.15$ K, $p_0 = 7.5 \times 10^6$ Pa, and slopes $a_T = -5 \times 10^{-4}$ K ^{−1} , $a_p = 3.2 \times 10^{-10}$ Pa ^{−1} .
<i>Medium-scale thermal mixing</i>			
thermal_conductivity (EffectiveThermalCon- ductivityPorosityMix- ing)	$k_{\text{eff}}(S)$	W m ^{−1} K ^{−1}	Effective thermal conductivity of the porous medium, with liquid and solid only (gas neglected): $k_{\text{eff}}(S) = nS k_\ell + (1 - n) k_s.$ The porosity n is given below, and k_ℓ, k_s are as above.
<i>Hydraulic constitutive laws (van Genuchten)</i>			
relative_permeabilit- y (RelativePermeability- VanGenuchten)	$k_{\text{rel}}(S)$	–	van Genuchten–Mualem relative permeability function with residual saturations $S_{\ell r} = 0.1$, $S_{gr} = 0$, exponent $m = 0.45$, and minimum liquid relative permeability $k_{\text{rel}}^{\text{min}} = 10^{-25}$: $S_e = \frac{S_\ell - S_{\ell r}}{1 - S_{\ell r} - S_{gr}},$ $k_{\text{rel}}(S_\ell) = S_e^{1/2} \left[1 - (1 - S_e^{1/m})^m \right]^2.$
saturation (Satura- tionVanGenuchten)	$S_\ell(p_c)$	–	Here S_ℓ is the liquid saturation. Water-retention curve (drying branch) in van Genuchten form with $S_{\ell r} = 0.1$, $S_{gr} = 0$, exponent $m = 0.45$ and air-entry pressure $p_b = 10$ MPa: $S_\ell(p_c) = S_{\ell r} + (1 - S_{\ell r} - S_{gr}) \left[1 + \left(\frac{p_c}{p_b} \right)^{1/(1-m)} \right]^{-m},$ where p_c is the capillary pressure.

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Table 3: Fixed material and fluid parameters and constitutive laws (continued).

XML name / property	Symbol	Unit	Value / definition
bishops_effective_stress (BishopsPowerLaw) <i>Poromechanical coupling and porosity (fixed in this study)</i>	$b(S)$	–	Bishop effective-stress function $b(S) = S^q$ with exponent $q = 1$, i.e. $b(S) = S$.
phi (porosity)	n	–	Total (water-loss) porosity used in the Richards and thermal mixing terms: $n = 0.105$.
biot (biot_coefficient)	α	–	In the present configuration this value is taken as fixed and is only used as a reference for the prior in Table 4. Biot coefficient, here fixed to $\alpha = 1.0$.

Table 4: Candidate hydraulic parameters and prior distributions for staged Bayesian calibration. Only the intrinsic permeability tensor field is implemented in the current workflow; scalar porosity and retention parameters should be released only after OGS state-output residuals are active.

Parameter / function	XML reference value	Prior distribution	Role if released
<i>Porosity and water-retention curve (later-release scalar parameters)</i>			
Porosity of Opalinus Clay	$n_0 = 0.105$	Normal: $n \sim \mathcal{N}(0.105, 0.02^2)$	Scalar porosity controlling storage and thermal mixing. Mean set to the XML value, standard deviation chosen to cover the range $n \approx 0.09$ – 0.17 reported for Mont Terri OPA: Nagra diffusion and rock-capacity data give HTO-accessible porosities of about 0.14 – 0.17 , with some method- and tracer-specific values around 0.09 – 0.11 [23, pp. 26–27, Tables 8–9 and pp. 44, 65–66, Tables 12 and 19], while the geomechanical compilation reports total porosity values in the same order [1, pp. 153–156]).
van Genuchten exponent (retention) m	$m_0 = 0.45$	Normal: $m \sim \mathcal{N}(0.45, 0.05^2)$	Scalar shape parameter of the water-retention curve. The mean is the XML value, consistent with laboratory WRC fits for Opalinus Clay where m typically lies in the range 0.4 – 0.6 depending on facies and stress path [24, pp. 22–30, Sect. 4].
Air-entry pressure p_b in VG WRC	$p_{b,0} = 10$ MPa	Uniform: $p_b \sim \mathcal{U}(5 \text{ MPa}, 20 \text{ MPa})$	Controls the suction range over which desaturation occurs. The XML value 10 MPa is taken as central; the bounds reflect roughly a factor-two scatter reported in fitted WRCs for OPA under different conditions (e.g. [24, pp. 31–33, Sect. 5] and [29, p. 4448]).
<i>Intrinsic permeability tensor (random field + global anisotropy)</i>			

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Table 4: Candidate hydraulic parameters and prior distributions for staged Bayesian calibration (continued).

Parameter / function	XML reference value	Prior distribution	Role if released
Permeability magnitude (log-field)	$k_x = 1.00 \times 10^{-19}$	Lognormal RF: $\ln k_{\text{mag}} \sim \mathcal{N}(\mu_k, \sigma_k^2)$. $\mu_k = \ln(6.32 \times 10^{-20})$; $\sigma_k = 0.5$. The prior generator draws this field independently of $\mathbf{x}$ before spatial correlation is imposed.	Scalar permeability magnitude field entering the intrinsic permeability tensor $\mathbf{k}_{\text{perm}}(\mathbf{x})$. The mean is chosen as the geometric mean of the XML principal values 1.00×10^{-19} and $4.00 \times 10^{-20} \text{ m}^2$; $\sigma_k = 0.5$ (natural-log scale) corresponds to an order-of-magnitude prior range consistent with OPA hydraulic characterisations [13, pp. 130–135] [6, pp. 3–5] [7, pp. 223–227] [29, pp. 4450–4452].
	$k_y = 4.00 \times 10^{-20}$		
	$k_{\text{ref}} = 6.32 \times 10^{-20} \text{ m}^2$		
Global anisotropy ratio $r = k_{\parallel}/k_{\perp}$	$k_x = 1.00 \times 10^{-19}$ $k_y = 4.00 \times 10^{-20}$ $r_0 = 2.5$	Lognormal (scalar): $\ln r \sim \mathcal{N}(\mu_r, \sigma_r^2)$. $\mu_r = \ln(2.5)$; $\sigma_r = 0.4$.	Global (spatially constant) anisotropy ratio between the permeability components parallel and normal to bedding. The prior mean reproduces the XML ratio; the chosen σ_r allows r in the interval of a few to about ten, which is compatible with directional anisotropies reported for Opalinus Clay [13, pp. 130–135] [7, pp. 225–227].
Global bedding-orientation angle θ in the tunnel cross-section	The current XML tensor $\mathbf{k}_{\text{i}}$ is aligned with the global axes, which we interpret as a reference angle $\theta_0 = 0 \text{ rad}$	Normal (scalar): $\theta \sim \mathcal{N}(0, \sigma_{\theta}^2)$, $\sigma_{\theta} = \pi/12$, i.e. a standard deviation of 15° .	Global (spatially constant) rotation of the intrinsic permeability tensor in the tunnel cross-section. The prior mean corresponds to the current XML configuration (no rotation); $\sigma_{\theta} = 15^\circ$ reflects moderate uncertainty in the effective alignment of hydraulic anisotropy with the structural bedding orientation as inferred from geological and geophysical observations [13, pp. 130–135] [6, pp. 2–5] [29, pp. 4447–4450].

In summary, the fixed set (Table 3) contains thermal properties, fluid properties, the Biot coefficient, the Bishop function, the detailed form of the saturation-dependent swelling strain, and the orthotropic elastic constants, which are constrained independently by thermal and mechanical experiments on Opalinus Clay (e.g. heating tests HE-B/HE-D/HE-E and geomechanical campaigns [9, pp. 270–279] [1, pp. 153–166] [8, pp. 211–222]).

The staged Bayesian workflow should start with the intrinsic permeability tensor $\mathbf{k}_{\text{perm}}$ because it is the only parameter field with a direct measurement target layer in the current files. It is parameterised as

$$\mathbf{k}_{\text{perm}}(\mathbf{x}) = k_{\text{mag}}(\mathbf{x}) \mathbf{R}(\theta)^\top \begin{pmatrix} r & 0 \\ 0 & 1 \end{pmatrix} \mathbf{R}(\theta),$$

where $k_{\text{mag}}(\mathbf{x})$ is a lognormal scalar random field and r and θ are global (spatially constant) scalar parameters describing, respectively, the anisotropy ratio and the bedding-related orientation of the tensor. This reflects the empirical evidence that permeability magnitudes in Opalinus Clay vary by roughly one to two orders of magnitude at the metre scale and that the principal directions of anisotropy are controlled by the large-scale bedding fabric and can be treated as homogeneous at the tunnel scale [13, pp. 130–135] [6, pp. 2–5] [7, pp. 225–227] [29, pp. 4447–4450]. The Gaussian priors for n and m , and the uniform prior for p_b , are retained as later scalar release options. They are centred at or around the XML values, which are themselves consistent with independent porosity and WRC measurements for Mont Terri samples [23, pp. 44 and 65–66, Tables 12 and 19] [24, pp. 22–33, Sects. 4–5]. They should remain fixed in the first production inversion unless NMR, ERT, RH, and Taupe/TDR state residuals show that permeability alone cannot explain the hydraulic evolution.

9 Available measurements and how they can enter the frozen OGS model

This section is based on the measurement catalogue in `../cda_knowledge_base/measurements`. The purpose is not to change the provided OGS model, but to define defensible observation and parameter mappings for inversion. In the current XML the unknowns solved by OGS are liquid pressure p , temperature T , and displacement $\mathbf{u}$. The secondary fields exported by the TRM process include velocity, saturation, stress, strain, and swelling stress. Measurements must therefore enter through one of three routes:

1. **State observation:** compare a measurement-derived quantity with an OGS output field after an observation operator, e.g. $S_\ell \mapsto \theta \mapsto \rho$ for ERT.
2. **Boundary-condition reconstruction:** convert atmospheric or niche measurements, especially RH, into liquid-pressure boundary curves.
3. **Parameter-field inference:** infer a heterogeneous and anisotropic intrinsic permeability tensor $\mathbf{k}(\mathbf{x})$, and possibly porosity or retention parameters, then regenerate the XML/mesh parameter input for each inversion sample.

The strongest first target is the intrinsic permeability tensor. The current model uses the constant diagonal parameter

$$\mathbf{k.i} = \begin{pmatrix} 1.00 \times 10^{-19} & 0 \\ 0 & 0.40 \times 10^{-19} \end{pmatrix} \text{ m}^2.$$

For the inversion this should be interpreted as a reference tensor only. A practical field parameterisation is

$$\mathbf{k}(\mathbf{x}) = \mathbf{R}(\theta)^\top \begin{pmatrix} k_{\parallel}(\mathbf{x}) & 0 \\ 0 & k_{\perp}(\mathbf{x}) \end{pmatrix} \mathbf{R}(\theta), \quad k_{\parallel}(\mathbf{x}) = r(\mathbf{x})k_{\perp}(\mathbf{x}),$$

with bedding-related orientation θ . The measurement chapter below explains which data constrain this tensor directly and which constrain it only indirectly through saturation, pressure, or water-content dynamics.

9.1 Measurement inventory by model relevance

The collected measurement catalogue is outside this report folder at `../cda_knowledge_base/measurements/`. Table paths are relative to that catalogue root. A machine-checkable version of the same inventory is kept in `inversion_workflow/observation_manifest.json`; its validator currently checks 28 local file/table/mesh conditions across the nine observation groups with zero failures. A second, model-facing layer is kept in `inversion_workflow/processed_observations/`. It contains normalized CSV tables generated from the collected workbooks and ZIP archives by `inversion_workflow/scripts/build_processed_observations.py`; every row keeps source-file and sheet/member provenance so values can be traced back to the raw emails or TeamBeam transfers.

Table 5: Measurement folders, physical quantity, and intended model-entry route.

Measurement	Files copied in the catalogue	Physical quantity	Model-entry route
ERT	<code>ert/source_files</code> : open-niche VTK ZIP, raw <code>tx0/ohm</code> , electrode positions, resistivity–water-content sheet/PDF, ERT slides.	Electrical resistivity or resistivity change; water content only after empirical conversion.	Observation operator from OGS saturation/water content to resistivity, preferably on the ERT inversion mesh or a projected common mesh.
NMR	<code>nmr/source_files</code> : weekly <code>.dat</code> files, seasonal NMR ZIP and extracted files, figures, NMR 2025/2026 presentations, coordinates.	Hydrogen/NMR-derived volumetric water content and relaxation-time information at sparse positions.	Point/line observation of water content after deciding whether the target is total water, mobile water, or a corrected free-water content.
Permeability pulse tests	<code>permeability_pulse_tests/source_files</code> : 2021/2024/2025 workbooks and characterization papers.	Interval-scale hydraulic/gas-test permeability or transmissibility around borehole sections.	Direct constraint on effective directional components or averages of $\mathbf{k}(\mathbf{x})$; not a direct cell-wise tensor value.

Table 5: Measurement folders, physical quantity, and intended model-entry route (continued).

Measurement	Files copied in the catalogue	Physical quantity	Model-entry route
Taupe/TDR	<code>taupe_tdr/source_files:</code> Taupe-WC.xlsx, coordinates, borehole figures, ISU/TD slides.	Apparent relative dielectric permittivity / TDR water-content proxy in EDZ distance bands.	Baseline-normalized saturation/water-content trend diagnostic; absolute residuals require confirmed Taupe unit/calibration.
Suction/RH	<code>suction_relative_humidity/ source_files:</code> OT_RH5--8.xlsx, location figure, annual report, TD slides/minutes.	Relative humidity and suction/water potential in boreholes and niches.	Boundary-condition reconstruction through Kelvin-equation liquid pressure, and validation of desaturation state through the retention curve.
Coord./layout	<code>coordinates_geometry_layou t/source_files:</code> measurement coordinate workbooks, figures, Tecplot layout.	Measurement locations, borehole labels, and model coordinates.	Mandatory geometric mapping for all observation operators.
Projection inputs	<code>model_projection_inputs/so urce_files:</code> projection script, projected VTU mesh, project/XML variant, 2D model archives, timestep notes.	Mesh and OGS-input workflow material, not a physical measurement.	Concrete bridge for mesh-based parameter fields and for aligning measurement outputs with OGS timesteps.
Bedding/geology	<code>bedding_geology_structure/ source_files:</code> bedding image, bedding page, characterization papers.	Bedding and fault/structure orientation, not a state variable.	Orientation prior for anisotropic $\mathbf{k}$; structural explanation for local ERT and pulse-test anomalies.
Other HM monitoring	<code>other_hm_monitoring/source _files:</code> TD minutes, modelling slides, levelling slides, HERMES input note, Tecplot layout.	Extensometer, mini-piezometer, crackmeter, laser scan, levelling, RH/T context.	Structured layout, levelling summary, and qualitative pressure/deformation validation; Geoscope and laser-scan raw exports still missing for hard residuals.

9.2 Source-file content pass and traceability

The catalogue has also been checked by a second pass through the copied files themselves, not only through the email summaries. The generated index `../cda_knowledge_base/measurements/measurement_content_deep_dive.md` contains 101 mined fact rows across the nine measurement folders, and each measurement folder has a local `DATA_CON
TENT_SUMMARY.md` plus `derived_files/content_deep_dive/content_summary.csv`. The mirrored navigation copy is under `../cda_knowledge_base/measurements_info`. This pass mines the workbook sheets, raw `.dat/.ohm/.tx0` files, ZIP member catalogues, and generated PDF/PPTX text extracts. Table 6 records the main data facts that should be treated as the report-level source checklist.

Table 6: Second-pass mined source-file facts used as a report traceability checklist.

Folder	Concrete copied data content	Modelling implication
ERT	15 copied source files; the TeamBeam ERT ZIP has 1,736 indexed members (1,675 <code>.vtk</code> fields and 61 timestep/list files) in dated folders from November 2019 to December 2024. The Thunderbird raw example has a 6,128-line <code>tx0</code> file, 2,656-line open <code>ohm</code> file, 2,524-line closed <code>ohm</code> file, and 72-electrode open/closed text files.	ERT can be audited down to raw inversion and electrode files, but it remains an indirect log-resistivity state observation. The ZIP is intentionally retained compressed while every member is indexed.

Table 6: Second-pass mined source-file facts used as a report traceability checklist (continued).

Folder	Concrete copied data content	Modelling implication
NMR	54 copied source/data files. The weekly 4S table has 21 rows from 2021-10-12 to 2022-03-20 with water content 9.897–13.169 vol.% and mean 12.004 vol.%. The weekly 4E table has 149 rows from 2021-10-06 to 2025-09-09 with water content 8.509–12.229 vol.% and mean 10.592 vol.%. The seasonal archive contributes 22 campaign tables and 294 position rows across Niches 3 and 4, with water content 2.766–22.810 vol.%.	NMR is the strongest local water-content evidence, but the source pass also makes the bound/interlayer-water problem visible: these values are total NMR water-content observations and cannot be equated mechanically to nS_ℓ .
Permeability pulse tests	6 copied source files, including two permeability workbooks. The workbook pass indexes chart sheets, interpreted 2021/2024 sheets, and raw pressure-decay sheets with thousands of non-empty cells; the processed observation layer extracts 204 interpreted interval rows and 6,687 pressure-decay rows.	This is the direct parameter-observation stream for $\mathbf{k}$, but the gas pulse tests are scalar finite-interval observations with uncertainty, not cell-wise tensor components and not liquid relative permeability.
Taupe/TDR	10 copied source files. Taupe_WC.xlsx has sheets A3, A4, A7, and A8, each with 212 dated rows from 2019-12-01 to 2025-10-10 and EDZ-band columns. The sheet values span approximately 9.87–11.00 in the copied workbook units over the four sheets.	Taupe/TDR has enough local table structure for trend diagnostics by sensor and EDZ band, but the absolute unit/calibration gate remains open; the report should not present these values as calibrated saturation.
Suction/RH	10 copied source files. The OT.RH5–8 workbooks contain daily two-column RH time series: RH5 has 1,042 data rows (94.081–95.993%), RH6 has 1,064 rows (93.323–94.988%), RH7 has 1,071 rows (13.188–97.272%), and RH8 has 1,070 rows (13.203–96.219%).	RH5/RH6 are the cleanest copied open-twin RH streams. RH7/RH8 low values are explicit source-level outliers or failure candidates, so RH-derived pressure boundaries must include sensor screening and provenance.
Coord./layout	8 copied source files. The September coordinate workbook has 109 rows and 10 columns; the November coordinate workbook has 39 rows and 8 columns. Both are indexed with numeric coordinate ranges and linked to the copied figures and VisualisationCDA.dat .	All point, line, interval, and borehole observations must pass through this geometry layer before they are compared with the current Niche-4 OGS mesh.
Bedding/geology	4 copied source files, including the bedding image, bedding-page PDF, and local characterisation-paper copies.	These files provide the structural prior for permeability anisotropy and local fault/EDZ interpretation, not a direct residual.
Other HM monitoring	12 copied source files. The content pass indexes five PDFs, three PPTX decks, the Tecplot-style VisualisationCDA.dat , and response-note placeholders for missing numeric exports and uncertainty information.	The current report can use these files for qualitative HM context and levelling summary evidence, but hard Geoscope, laser-scan, crackmeter, and related residuals still require numeric exports and uncertainty definitions.

Table 6: Second-pass mined source-file facts used as a report traceability checklist (continued).

Folder	Concrete copied data content	Modelling implication
Projection inputs	87 copied source files, including four ZIP archives, 57 XML files, 15 VTU files, the projection script, project files, timestep slides, and the extracted report ZIP. The source pass confirms the projection workflow references <code>generate_projections.py</code> , <code>specifications.pptx</code> , and <code>cd_a_open_niche_quad.prj</code> .	This folder is the implementation bridge from measurements to OGS: it defines the mesh fields, projection support, timestep convention, and exchanged model versions without changing the frozen source model equations.

9.3 ERT: resistivity, water content, and saturation

The ERT archive contains a long open-niche time series: 1,736 files, including 1,675 VTK files and 61 monthly/timestep lists from November 2019 to December 2024. The processed tables `ert_archive_inventory.csv` and `ert_time_steps.csv` catalogue those ZIP members without unpacking the full archive. They list 1,691 timestep-list entries and explicitly flag 16 entries without a matching VTK member, so an observation operator should not assume a one-to-one date-VTK map without this check. The companion processed files `ert_nmr_resistivity_pairs.csv` and `ert_kruschwitz2004_relation.csv` normalize the local water-content/resistivity calibration workbook. The raw ERT example recovered from Thunderbird contains one `tx0` file with both niches and separate `ohm` inversion input files for the open and closed niches. Gesa’s accompanying note states that the raw `tx0` file contains open-niche measurements #1–#2592 and closed-niche measurements #2593–#5184.

ERT does not measure water content or saturation directly. It measures an electrical response that is inverted to apparent/effective resistivity. Zieffle et al. use the CD-A NMR/ERT crossplot to establish an empirical relation between electrical resistivity and volumetric water content; the paper writes an Archie-type relation $\rho = c\rho_{\text{fluid}}\theta^{-m}$ and then uses the fitted relationship for comparison with simulated water content [29, p. 4453, Eq. 4 and p. 4457, Sect. 5.1]. This is consistent with Archie’s original use of formation factor and saturation-resistivity relations in clean porous media [2, pp. 54–58]. However, CD-A is claystone, not clean sandstone: clay-bearing systems can carry current both through the pore fluid and along mineral–water surfaces, with cation-exchange/surface-conductivity terms and saturation transforms entering the conductivity model [16, pp. 23925–23927 and p. 23930, Eq. 17]. The CD-A ERT calibration must therefore remain local and empirical rather than a universal saturation transform.

For modelling, the correct direction is:

$$S_{\ell}^{\text{OGS}}(\mathbf{x}, t) \rightarrow \theta_{\text{model}}(\mathbf{x}, t) = n S_{\ell}(\mathbf{x}, t) \rightarrow \rho_{\text{pred}}(\mathbf{x}, t) \rightarrow \mathcal{O}_{\text{ERT}}[\rho_{\text{pred}}],$$

where $\mathcal{O}_{\text{ERT}}$ includes projection/interpolation to the ERT mesh and any depth/ring averaging. The inverse direction, converting ERT to “measured saturation”, is useful for plots but should not be treated as raw truth because fractures, bedding-parallel conductive paths, electrode geometry, inversion smoothing, and the different ERT/OGS meshes all enter the observation. The September 2025 slides explicitly list these as open inverse-problem questions, especially filtering unstable points and handling the mismatch between the ERT computational domain and the OGS domain [26, PDF pp. 3, 16–17].

The workflow now makes the calibration part of $\mathcal{O}_{\text{ERT}}$ explicit in `inversion_workflow/scripts/build_ert_observation_operator.py`. The script fits the paired NMR/resistivity rows and the smooth workbook relation columns and writes `ert_water_content_resistivity_operator.csv`. The current default first-test relation is the workbook `model_data2019` curve,

$$\rho_{\Omega\text{m}} = 1.108 \theta^{-1.58}, \quad \theta = n S_{\ell},$$

over the calibrated water-content range $0.01 \leq \theta \leq 0.18$. The direct Niche-4 paired NMR/resistivity fit is retained as a diagnostic, but its scatter is larger (0.202 log10 resistivity units), so it should not be mistaken for a completed ERT forward model.

The spatial part of the ERT operator is now represented separately by `build_ert_spatial_projection_lookup.py`. It reads a reference ERT VTK from `ERT_meas_Niche_open.zip`, records the raw ERT inversion mesh (3,126 points and 5,966 triangular cells), applies the explicit provisional transform

$$x_{\text{model}} = x_{\text{ERT}}, \quad y_{\text{model}} = z_{\text{ERT}} + 500 \text{ m},$$

and maps ERT cell centroids to the OGS bulk mesh. The generated `ert_spatial_projection_lookup.csv` has 4,676 ERT cells inside an OGS cell and 2,035 cells that are also inside the approximate 1.5 m central support mask noted in the local slides [26, PDF pp. 16–18]. This turns the ERT mesh mismatch into a reproducible lookup instead of an undocumented manual step. However, the transform and the exact “35 cm in rock” near-niche support mask still need

confirmation against the agreed ERT/FEM coordinate convention before ERT log-resistivity residuals are activated. The generated ERT semantics audit `inversion_workflow/processed_observations/ert_measurement_semantics.md` records the same gates in row-level form. It keeps 1,691 ERT timestep rows, of which 1,675 have matching VTK members and 16 are blocked by missing VTK files. The reference projection lookup has 5,966 ERT cells; 4,676 are inside an OGS cell, 2,194 are inside the approximate 1.5 m support screen, and 2,035 satisfy both conditions. These 2,035 rows are the current provisional ERT support under the explicit transform/support assumptions. The audit also records the recommended residual as $\log_{10} \rho_{\text{pred}} - \log_{10} \rho_{\text{ERT}}$ and explicitly excludes use of ERT as measured saturation, measured water content, or measured permeability.

The run-local diagnostic `inversion_workflow/runs/direct_fit_observation_run/ert_resistivity_diagnostic.md` now performs that comparison for the direct reference run. It samples $\theta = nS_\ell$ from the OGS outputs on the 2,035 provisional ERT-support cells, converts the sampled water content with the default Kruschwitz relation, and compares the prediction with the nearest ERT inversion field within a 10-day tolerance. The diagnostic compares 162,800 cell-time rows across 80 OGS output times; 3 output times are outside the time tolerance. Its area-weighted residual $\log_{10} \rho_{\text{pred}} - \log_{10} \rho_{\text{ERT}}$ has MAE 0.254 and RMSE 0.300. This closes the local OGS-output sampling step, but it is still a diagnostic rather than a likelihood term because the coordinate transform, exact near-niche support mask, and ERT inversion uncertainty/correlation model are not yet accepted.

The cross-run audit `build_ert_candidate_discrimination_audit.py` then applies the same compact diagnostic to 66 executed OGS runs with output fields. Each run compares 162,800 ERT cell-time rows over 80 output times. Across the current candidate family the area-weighted ERT MAE spans 0.019635573360798686 log10 units. The ERT-only best run is `broad_continuous_001_001_length_0p023m_shift_1p004`, with MAE 0.25407209193077057; the best active-objective run is `local_basis_sampler_002_basis_024_det_1_0p0075_s_1p000`, with ERT MAE 0.2541796182834413. The combined-objective/ERT-MAE correlation is 0.8894403368395667, so candidates that fit the current active permeability+NMR objective better generally do not improve the provisional ERT residual. This is candidate-screening evidence only; it does not remove the transform, support-mask, or ERT uncertainty gates.

The support-mask part of that gate is now quantified locally by `build_ert_support_sensitivity_audit.py`. The audit reuses the row-level ERT diagnostic on six representative executed runs and restricts the current provisional 1.5 m support to tighter radial caps and inner/outer annuli. It evaluates nine support variants. The ERT-only run `broad_continuous_001_001_length_0p023m_shift_1p004` remains the best mean support-rank candidate, with mean rank 1.67 and worst rank 4.0, but the tightest or outer support subsets can swap the winner to `local_basis_sampler_003_basis_029_det_1_0p0100_s_1p000`. This strengthens the diagnostic record because support sensitivity is now explicit; it still does not authorize ERT likelihood weights until the agreed tunnel-contour support and uncertainty/correlation model are fixed.

ERT constrains permeability indirectly: permeability changes the simulated desaturation/resaturation dynamics, which then affects predicted resistivity. The ERT data are therefore valuable for heterogeneous $\mathbf{k}(\mathbf{x})$, but only through a forward observation model; they are not a direct permeability measurement.

9.4 NMR: total water signal versus OGS liquid saturation

The NMR folder contains two weekly monitoring tables and a seasonal campaign archive. The weekly 4S table has 21 rows from October 2021 to March 2022; the weekly 4E table has 149 rows from October 2021 to September 2025. The values are reported as volumetric water content with confidence information and relaxation-time columns. The seasonal ZIP contains 22 .dat data files and 22 figures, covering Niche 3 and Niche 4 campaigns between 2019 and 2025. The processed tables `nmr_weekly.csv` and `nmr_seasonal_profiles.csv` combine these into 170 weekly rows and 294 seasonal campaign-position rows. The weekly table flags the February–April 2025 4E detuning caveat directly in the row-level `caveat` column.

The CD-A paper treats NMR as the more direct water-content measurement used to anchor the ERT interpretation; Fig. 6 shows the volumetric water content measured by single-sided NMR at repeated surface positions, and Fig. 7 uses the NMR/ERT crossplot for the empirical resistivity-water-content conversion [29, pp. 4451–4453]. That does not automatically mean that NMR equals OGS saturation. NMR observes hydrogen-bearing water with relaxation behaviour controlled by pore geometry, surface relaxation, and clay effects. The CD-A NMR 2026 presentation in the local archive specifically flags interlayer/bound water as a live interpretation issue; general NMR petrophysics literature also treats clay effects and T2 distributions as a key limitation when connecting NMR to capillary pressure and producible/mobile water [4, PDF pp. 23–24][10, pp. 761–763]. An open-access dual-scale NMR modelling study makes the same point from the forward modelling side: T2 distributions are used to infer pore size and permeability, but their response is affected by internal magnetic-field gradients and unresolved clay micro-porosity, so clay-rich regions require either explicit microstructure or effective parameters rather than a direct saturation reading [5, pp. 453–455].

The defensible modelling map is therefore:

$$\theta_{\text{NMR}}^{\text{obs}} = \theta_{\text{mobile}} + \theta_{\text{bound/interlayer}} + \epsilon_{\text{NMR}}, \quad \theta_{\text{mobile}}^{\text{model}} \approx nS_\ell.$$

The generated bound-water sensitivity audit in `inversion_workflow/processed_observations/nmr_bound_water_sensitivity.md` quantifies how large this caveat is for the current model. With fixed $n = 0.105$, 315 of 464 NMR target

rows exceed the maximum possible model-side $\theta = nS_\ell$ if interpreted as mobile water without correction. Within the currently usable Niche-4 mapped subset, 162 of 287 rows exceed n . The required positive bound-water/free-water offset for usable rows has $p95 = 0.0402$ and maximum = 0.1231; a tested uniform subtraction of 0.05 maximizes the simple physical-row count but still leaves seven non-physical usable rows. This means a single global subtraction is not a clean correction.

The cross-run NMR candidate bias audit `build_nmr_candidate_bias_sensitivity_audit.py` tests the ranking consequence of that caveat across 66 executed OGS runs with 192 active NMR rows each. Under the current raw absolute- θ objective, the best run is `local_basis_sampler_002_basis_024_det_1_0p0075_s_1p000`, with combined objective 3156.353066948979. If each NMR label is allowed a non-negative constant bound/interlayer-water bias, or if only within-label anomalies are compared so that constant offsets cancel, the best diagnostic run becomes `broad_continuous_001_003_length_0p021m`, with diagnostic combined objective 505.614305828437. The rank correlation between the current raw objective and the per-label-bias diagnostic is 0.6351529067946979. Therefore the present permeability ranking is conditional on the unresolved NMR free-water interpretation, not a final NMR-robust inversion result.

The provisional NMR objective decision in `inversion_workflow/nmr_objective_decision.md` therefore recommends a within-label trend/anomaly residual as the first final NMR likelihood candidate, while the internal decision register records a current not-default policy for the active report state. Thus the raw-theta active objective is left unchanged, and the trend/anomaly residual is used as explicit scenario/provisional-likelihood evidence unless the modelling team reopens the default-objective decision. The residual compares

$$(\theta_{\text{model}} - \overline{\theta_{\text{model}}}) - (\theta_{\text{NMR}}^{\text{obs}} - \overline{\theta_{\text{NMR}}^{\text{obs}}})$$

within each observation-family/measurement-label group ℓ . Constant bound/interlayer-water and campaign offsets then cancel to first order, so the NMR stream constrains water-content changes rather than absolute mobile water content. This residual now has an executable assembler mode, `--state-objective-mode nmr_within_label_trend_anomaly`, and the separate `inversion_workflow/nmr_trend_anomaly_active_objective.md` ranking validates it against the diagnostic audit to numerical roundoff. In the 66 full active direct-permeability plus NMR runs, it selects `broad_continuous_001_003_length_0p021m` with objective 505.614305828437; the raw-objective incumbent is rank 14 under this residual. The final NMR residual-policy gate in `inversion_workflow/nmr_final_residual_policy_gate.md` records the resulting promotion boundary: no final NMR policy is selected yet, raw absolute theta remains the current-report default with caveats, within-label trend/anomaly is the preferred provisional final-policy candidate, and the follow-up planner recommends pausing new OGS batches for that promoted-NMR mode because the best unevaluated direct advantage is below the materiality threshold. The matching acceptance-record template `inversion_workflow/nmr_final_residual_policy_acceptance_record_template.md` is the signoff guardrail for that choice. It contains the four NMR residual options, currently records zero of one required primary approvals, is not ready to apply, records no actual decision, changes no active objective, promotes no field, and recommends no new OGS batch. The package explicitly rejects a single global NMR offset as a final correction unless collaborators provide a physical justification, and keeps porosity fixed because NMR water content alone cannot separate porosity, saturation, and bound water.

For inversion, NMR should initially be used as a calibration/validation anchor for near-surface water-content trends and for checking the ERT conversion. If a free-water correction becomes available from Stephan's lab interpretation, the NMR likelihood can be tightened. Without that correction, use a bias/error term or compare changes $\Delta\theta$ rather than absolute saturation.

Important local caveats from the email archive:

- Columns 4–5 of the weekly files can be ignored for modelling.
- Technical failures cause gaps.
- February–April 2025 at 4E likely overestimates water content by about 1 vol.% because the device was detuned to the wrong frequency.
- Seasonal data before autumn 2021 are floor/ceiling only; September 2020 is floor only; winter 2024 is missing.

9.5 Permeability pulse tests: scalar interval data versus tensor field

The permeability folder contains two main workbooks: `2025-09-05_CD-A_Permeability.xlsx` and `Permeability_CDA_all_2025.xlsx`. The latter includes 2021 data from BCDA.24/BCDA.25 and 2024 data from BCDA.32/BCDA.33. The September slides describe modified COMDRILL double-piston packer pulse tests with a 10 cm test interval, nitrogen injection up to 1 bar, and pressure-decay monitoring [26, PDF p. 5]. The reported value is an interval-scale intrinsic-permeability estimate for a 3D volume around the tested borehole segment, with experimental/evaluation uncertainty of about half an order of magnitude. The processed files `permeability_interpreted_values.csv` and `permeability_pressure_decay.csv` extract 204 interpreted interval rows and 6,687 raw pressure-decay rows while preserving source workbook, sheet, row, campaign year, borehole block, depth, permeability, transmissibility, and pressure-time provenance.

This is the most direct measurement class for $\mathbf{k}$, but it still does not give a cell-wise tensor. It is an effective scalar or directional response of a finite test volume and borehole orientation. The CD-A characterization paper reports that pulse tests indicate a stronger EDZ and permeability increase in the open twin than in the closed twin, and later discusses interval permeability values around 10^{-16} to 10^{-13} m² depending on niche, depth, and anomalies [28, PDF pp. 1, 11, Sect. 4.3 and Sect. 6]. The workbooks and slides further distinguish 2021 (about 1.5 years after excavation) and 2024 (about 4.5 years after excavation).

The nitrogen-gas detail is crucial. The OGS TRM Darcy law uses liquid intrinsic permeability multiplied by liquid relative permeability:

$$\mathbf{v}^l = -\frac{\mathbf{k}_{\text{rel}}(S_\ell)}{\mu_w} \nabla p.$$

Gas pulse tests can estimate the medium’s intrinsic permeability only after accounting for gas slippage and test conditions. Klinkenberg showed that apparent gas permeability depends on reciprocal mean pressure and that extrapolation to infinite pressure gives the liquid-equivalent permeability constant [11, pp. 200, 205–207]. In CD-A, this means a nitrogen pulse-test value should enter the inversion as a noisy effective observation of intrinsic permeability, not as liquid hydraulic conductivity and not as $k_{\text{rel}}\mathbf{k}$. This separation is also consistent with Opalinus Clay gas-transport literature: gas flow in low-permeability claystone depends on intrinsic permeability, porosity, gas pressure, water saturation, capillary threshold, and hydromechanical state; once gas entry is exceeded, mobility is controlled by intrinsic permeability together with relative-permeability and retention relations [13, pp. 124–125, Sects. 2.1–2.2]. The pulse-test interpretation therefore informs $\mathbf{k}$, while liquid relative permeability and retention remain separate OGS constitutive choices.

Recommended observation operator:

$$d_i^{\text{pulse}} \approx \frac{1}{|\Omega_i|} \int_{\Omega_i} \mathbf{e}_i^\top \mathbf{k}(\mathbf{x}) \mathbf{e}_i w_i(\mathbf{x}) d\mathbf{x} \quad \text{with lognormal error,}$$

where Ω_i is the interval influence volume, $\mathbf{e}_i$ is the dominant borehole/test direction or sensitivity direction, and w_i is an approximate sensitivity weight. If the direction/sensitivity is not reconstructed, use the pulse data as order-of-magnitude constraints on EDZ zones rather than point tensor entries.

The working folder now implements the first direct-target version of this idea in `inversion_workflow/scripts/build_permeability_observation_targets.py`. The script reads the interpreted permeability rows, the borehole line samples, and the mesh lookup table, then writes `permeability_observation_targets.csv` and `permeability_observation_cells.csv`, plus `permeability_missing_geometry_audit.md` for interpreted rows that have source orientation evidence but no labelled endpoints. It does not treat the scalar gas pulse-test result as a tensor component. Instead, it records a noisy interval-scale log-permeability target with cell weights over a nominal 10 cm along-borehole interval. The current status is: 75 rows are usable as initial direct constraints in the local OGS mesh, 27 rows map outside the current mesh and are retained but excluded from the first fit, 98 older rows have interpreted values but lack matching borehole endpoint geometry, and 4 rows have no positive interpreted permeability. The 98 unmapped rows are grouped into five audit entries: BCD-A24 and BCD-A26 are retained as vertical tests, BCD-A25 and BCD-A27 as horizontal tests, and BFM-D19 as nearly horizontal evapometer data, but none is activated until labelled endpoint geometry or an accepted digitized trace can project the interval into OGS cells. The companion request package `inversion_workflow/processed_observations/permeability_endpoint_geometry_request.md` turns this into a concrete follow-up item: it asks for labelled start/end coordinates, coordinate frame, depth-zero reference, along-borehole direction, and interval-position convention for BCD-A24, BCD-A25, BCD-A26, BCD-A27, and BFM-D19. The same builder writes a 98-row blocked-observation table, so the inactive values can be reactivated reproducibly if the endpoints or an approved digitized trace are provided. The generated semantics audit `inversion_workflow/processed_observations/permeability_measurement_semantics.md` keeps this decision machine-checkable: 204 interpreted rows are present, 200 have positive permeability, 75 are active direct candidates, 27 are outside the current mesh, and 98 are blocked only by missing endpoint geometry. The active subset comes from BCD-A32 and BCD-A33, with usable log10 permeability values spanning -21.0 to -12.10 m². The audit explicitly labels every row as a nitrogen pulse-decay scalar interval observation of intrinsic permeability and records that it is not a measurement of hydraulic conductivity, saturation, liquid relative permeability, or a direct tensor component.

The companion evaluator `inversion_workflow/scripts/evaluate_permeability_targets.py` is the first machine-checkable likelihood component. It reads a generated mesh field `k_i_rd`, evaluates $\mathbf{e}^T \mathbf{k} \mathbf{e}$ on the selected interval cells, and compares predicted and observed permeability in log10 space. For the current `smoke_test` field, duplicate-aware weighting leaves an effective 52 direct targets, with a weighted RMSE of 2.70 log10 units. The best single global multiplier is only $10^{1.2285} \approx 16.9$, reducing the RMSE to 2.40 log10 units. This baseline confirms that the current generated field is only a prior/proposal field; the pulse-test data require heterogeneous local high-permeability intervals rather than a uniform rescaling.

A first diagnostic direct fit is implemented in `inversion_workflow/scripts/fit_permeability_field_from_targets.py`. It scales only the mesh cells selected by usable pulse-test intervals, preserves the existing tensor orientation and anisotropy, and stores the applied log10 multiplier as mesh-cell diagnostic fields. In `runs/direct_permeability_fit/`, this reduces the duplicate-aware direct-target RMSE from 2.70 to 1.61 log10 units by adjusting 30 cells, with applied log10 multipliers from -1.48 to $+5.80$. This is useful for locating forced high-permeability intervals, but it is not yet a full OGS inversion because it does not rerun the THM problem or include ERT, NMR, Taupe, or RH state observations.

The prior/proposal space has also been tested explicitly with `run_direct_permeability_prior_sweep.py`. A 12-candidate sweep generated heterogeneous anisotropic fields using the two usable pulse-test segment orientations and the bedding-prior angle as tensor-angle candidates. The best generated prior field has RMSE 2.41 log10 units and objective 605.00, with an optimal uniform multiplier of only 1.15; after that multiplier the RMSE remains 2.41. This improves on the original smoke-test prior but remains far worse than the direct cell-fit diagnostic. The conclusion is that the measured pulse-test intervals require local permeability structure, not only a different global scale or a generic random anisotropic prior.

A smoother interval-anchored field is implemented in `fit_smooth_permeability_field_from_targets.py`. This script takes the same duplicate-aware target-derived log10 multipliers used by the direct cell diagnostic, but spreads them from the 30 anchor cells through a finite-cutoff Gaussian kernel. It therefore keeps the tensor orientation and anisotropy while producing a neighbourhood of affected cells instead of isolated point corrections. The tested length scales show the expected tradeoff: 0.05 m affects 471 cells and gives RMSE 1.67 log10 units with objective 291.29, while 0.80 m affects 8,354 cells and relaxes to RMSE 2.13 with objective 470.95. An extended search now also varies shift damping. Its best direct candidate uses a 0.025 m smoothing length, affects 208 cells, and reaches RMSE 1.61 log10 units with objective 269.97. This is still smoother than the 30-cell direct diagnostic, but more localized than the earlier 0.05 m proposal. The candidate has been passed through the per-candidate driver as `candidate_smooth_0p025m_search_driver`; it is currently the best direct-fit field before state-observation comparison.

Because the direct objective alone favours the strongest local correction, the workflow now also ranks the smooth candidates with `rank_regularized_permeability_candidates.py`. The ranking is not a calibrated geological prior; it is a transparent selection aid that separates direct pulse-test misfit from the sum of squared cell-wise log10 permeability updates. The current Pareto tradeoff set contains three 0.025 m support fields: the full-shift `length_0p025m` field (objective 269.97, RMSE 1.61), the 0.75-shift field (objective 301.16, RMSE 1.70), and the 0.5-shift field (objective 392.72, RMSE 1.94). A weak field-update penalty still selects the full-shift field, a moderate penalty selects the 0.75-shift field, and a strong penalty selects the 0.5-shift field. At least the direct winner and one damped regularized candidate should be carried into the first OGS state-output comparison, so that the eventual inversion is not judged only on direct permeability overfit.

That handoff is now reproducible with `run_regularized_ogs_candidate_set.py`. The script reads the Pareto/scenario winners and prepares all three fields through the same candidate harness. The current executed set runs `length_0p025m`, `length_0p025m_shift_0p750`, and `length_0p025m_shift_0p500` through OGS with the collected 6.5.4 SIF via Dockerized Apptainer. Each run produced 83 output VTUs, 37,184 sampled state rows, and 192 active sampled NMR objective rows. The combined objectives are 3166.61, 3198.49, and 3289.65, respectively. The state objective is similar for the three fields (about 2897), so this first combined ranking is still driven mainly by the direct permeability term; it is an OGS-backed candidate comparison, not yet a final inversion sampler.

The next-batch decision is now recorded by `build_adaptive_combined_candidate_plan.py`. The first adaptive batch has now executed `length_0p050m`, `length_0p050m_shift_0p750`, and `length_0p075m` through the same OGS harness. Each run produced 83 output VTUs, 37,184 sampled state rows, and 192 active NMR objective rows. Their combined objectives are 3276.57, 3300.91, and 3353.88; none improves on the `length_0p025m` objective of 3166.61. The broader-support batch also increases the sampled NMR state objective, so the executed state term is no longer flat over that six-candidate set. A subsequent local-refinement grid around the 0.025 m support field executed `length_0p013m` and `length_0p019m`; both keep the same direct RMSE of 1.61 log10 units and improve the combined objective to 3159.66 and 3164.96, respectively. The first local-bracketing batch then executed `length_0p013m_shift_0p875`, `length_0p013m_shift_1p125`, and `length_0p031m`. Their combined objectives are 3167.34, 3167.21, and 3219.35, so neither shift perturbation nor the 0.031 m full-shift support improves on the 0.013 m local best. A finite-candidate Bayesian proposal then executed `length_0p019m_shift_1p125`, `length_0p019m_shift_0p875`, and `length_0p025m_shift_1p125`; their combined objectives are 3172.69, 3172.37, and 3173.39, again worse than the 0.013 m local best. A first continuous-proposal batch then executed `length_0p006m_shift_0p992`, `length_0p007m`, and `length_0p007m_shift_0p972`; their combined objectives are 3156.78, 3157.74, and 3158.16. A focused lower-support continuous batch then executed `length_0p004m_shift_0p995`, `length_0p003m_shift_0p986`, and `length_0p004m_shift_0p994`; their combined objectives are 3156.38, 3156.47, and 3156.38. A repeatable loop driver, `run_continuous_inversion_loop.py`, then refreshed the lower-support proposal, executed `length_0p003m_shift_1p006`, `length_0p005m_shift_0p985`, and `length_0p004m_shift_1p014`, and merged those rows into a cumulative lower-support evidence table. Their combined objectives are 3156.37, 3156.49, and 3156.44. A second loop then executed `length_0p004m_shift_0p992`, `length_0p004m_shift_1p020`, and `length_0p006m_shift_0p975`; their combined objectives are 3156.40, 3156.53, and 3156.70, so none improves the first-loop incumbent. The current best executed smooth field is therefore still `length_0p003m_shift_1p006`. A broad continuous evidence batch then executed `length_0p023m_shift_1p004`, `length_0p022m_shift_0p998`, and `length_0p021m`; their combined objectives are 3159.06, 3169.30, and 3169.31, so the broad uncertainty-ranked batch also did not improve the incumbent. The generalized loop driver then ran a repeatable broad iteration over `length_0p015m`, `length_0p016m_shift_0p968`, and `length_0p010m`; their combined objectives are 3161.36, 3165.33, and 3157.70, again leaving the incumbent unchanged. The regenerated adaptive planner treats 32 executed smooth-family candidates as evidence and proposes the next local batch led by `length_0p008m_shift_1p011`, `length_0p007m_shift_1p020`, and `length_0p007m_shift_1p022`. This is evidence that the lower-support continuous region has been useful, but the focused local family is now near a plateau and still not a substitute for a production optimizer or ensemble sampler.

The separate anisotropy sensitivity pass `build_anisotropy_sensitivity_plan.py` checks a different question: whether the direct pulse-test layer prefers a different global principal direction or anisotropy ratio than the current bedding-prior tensor. Starting from `length_0p003m_shift_1p006`, it preserves each cell's geometric-mean permeability and sweeps 35 angle/ratio combinations. The best direct candidate is the unmodified 144°, 2.5:1 field itself, with direct objective 269.83549 and zero improvement over the baseline; the next candidate, 140° with the same ratio, is already worse by 0.05086 objective units. This does not prove that anisotropy is known perfectly, but it means a global angle/ratio-only OGS batch is not a good use of computation. The later local-basis, local-anisotropy, cross-stream hybrid, structural/EDZ, and support lower-bound audits show that the next permeability search should wait for a support/likelihood decision or for currently gated streams such as ERT/Taupe/RH before adding broader tensor degrees of freedom.

The first richer local-basis diagnostic is recorded by `build_local_basis_sampler_plan.py`. It starts from the same `length_0p003m_shift_1p006` incumbent, keeps the governing OGS model frozen, and builds 30 residual-derived anchor cells from the active permeability pulse-test targets. The sampler scores 131 direct pulse-test candidates in which local coefficients can move independently around those anchors while only the run-local `k_i_rd` tensor magnitude field changes. The best direct row, `basis_004_det_1_0p0015_s_1p000`, improves the direct pulse-test objective from 269.83549 to 269.81806, with weighted RMSE changing from 1.610768 to 1.610715 log10 units and applied log10 increments between -0.03472 and 0.00885. This is a true but very small direct-layer improvement. The selected OGS batch contains three diverse local-basis candidates affecting 30, 37, and 47 cells. The batch has now been executed through the release-gated OGS harness. All three runs returned code 0, passed the release gate, sampled 192 active NMR state rows, and improved the combined incumbent slightly: `basis_024_det_1_0p0075_s_1p000` has combined objective 3156.35307, compared with 3156.37301 for the previous `length_0p003m_shift_1p006` incumbent. The small improvement makes it the current accepted best field, but it is still a local deterministic proposal result, not a converged inversion.

The local tensor-anisotropy screen then checks whether the next field family should release orientation or anisotropy ratio locally instead of continuing magnitude-only updates. The script `build_local_anisotropy_sampler_plan.py` starts from the accepted local-basis incumbent `basis_024_det_1_0p0075_s_1p000`, preserves each cell's geometric-mean permeability, and changes only the local principal direction and anisotropic split around the 30 mapped pulse-test anchor cells. It scores 80 direct pulse-test candidates across residual-sign, segment-parallel, segment-perpendicular, and local-isotropization modes. The best candidate, `local_anis_isotropize_10p015_s0p5_r1`, is worse than the incumbent: direct objective 270.00798 versus 269.81806, a +0.18992 objective increase. Thus the active direct permeability data do not justify a local tensor-orientation or anisotropy-ratio OGS batch at this stage.

The cross-family production convergence record is now recorded by `build_production_sampler_convergence_audit.py` and `runs/production_sampler_convergence/PRODUCTION_SAMPLER_CONVERGENCE.md`. This audit standardizes 54 accepted smooth-family OGS rows and 11 accepted local-basis rows, writes a stage-by-stage convergence history, and scores a 394-candidate smooth/local-basis pool with a random-forest surrogate for the sampled NMR state objective plus the known direct permeability objective. Five production handoff rounds have executed thirty candidates; all passed release gates, sampled 192 active NMR rows, and were worse than the incumbent. The best production row, `basis_023_det_1_0p0075_s_0p750`, has combined objective 3156.35476; the fifth round's best row, `length_0p004m_shift_1p031`, has combined objective 3156.78544, while the accepted local-basis incumbent remains `basis_024_det_1_0p0075_s_1p000` at 3156.35307. Of the 329 remaining unexecuted diagnostic rows, 210 have materialized mesh files and are eligible for a release-gated execution handoff; 119 random local-basis diagnostics are kept in the score table but excluded from the batch because they do not yet have mesh files. The refreshed top executable handoff row is `length_0p007m_shift_1p020`, with lower-confidence-bound objective 3156.94007 and a diagnostic probability of improvement of 0.0166. Because the local-basis improvement over the previous smooth incumbent is only 0.01994 objective units and the cross-validated state-objective RMSE of the surrogate is about 10.43, any next batch should be treated as convergence evidence, not as proof that the inversion is complete. The generated `PRODUCTION_SAMPLER_DECISION.md` therefore recommends `pause_active_production_sampling` for the current direct-permeability plus NMR objective: all five production rounds failed to improve the incumbent, the next top handoff probability of improvement is low, and all proposed lower-confidence bounds are above the incumbent. This pause is scoped only to the current active objective; it does not close the ERT, Taupe/TDR, RH, or later-parameter gates.

The older smooth-family Bayesian proposal layer also writes `runs/bayesian_candidate_proposal/OPTIMIZER_CANDIDATE_PROPOSAL.md`. This is a Bayesian surrogate over the already generated finite-grid and continuous smooth-field pool: it records the 32 executed smooth-family candidates as evidence, keeps all 32 represented in the generated-candidate score table, reconstructs fallback feature rows from executed search results where proposal directories had been overwritten, adds the known direct permeability objective for each unexecuted mesh, and ranks by a lower-confidence-bound acquisition. Its current top proposal is `length_0p003m_shift_0p972`; the previous broad top proposals have now been executed and proved worse than the incumbent. The artifact is best read as a transparent proposal diagnostic; a production inversion still needs an optimizer or ensemble sampler loop over accepted fields. The first continuous proposal layer is now recorded in `runs/continuous_bayesian_candidate_plan/CONTINUOUS_CANDIDATE_PLAN.md`. It samples support length and shift scale outside the fixed finite grid, writes 136 new run-local permeability meshes, evaluates the direct pulse-test objective for each, and uses the current generated-table evidence intersection to predict the NMR state-objective component. Its current proposed OGS batch, excluding already executed continuous rows, is `length_0p009m_shift_0p979`, `length_0p008m_shift_1p040`, and `length_0p010m_shift_1p050`. Those rows are not yet accepted evidence; they must be executed and sampled through the same release-gated OGS harness before comparing

them with the current `length_0p003m_shift_1p006` best field. The focused lower-support continuous plan separately proposes `length_0p003m_shift_0p972`, `length_0p004m_shift_1p031`, and `length_0p006m_shift_0p970`; after two executed lower-support loops its best probability of improvement is effectively zero, so this local plan is now mainly a plateau diagnostic.

9.6 Taupe/TDR: dielectric proxy and EDZ-band trends

The Taupe folder contains `Taupe.WC.xlsx` with sheets A3, A4, A7, and A8. Each sheet has 212 rows from 2019-12-01 to 2025-10-10 and columns for EDZ distance bands 0-50, 0-10, 10-20, 20-30, 30-40, and 40-50. The processed `taupe_tdr_bands.csv` table reshapes this workbook to 5,088 sensor-date-EDZ-band rows and keeps the workbook column name and EDZ-band limits in centimetres. The 2026 ISU slides identify the method as differential TDR using TAUPE sensor cables [12, PDF p. 1]. The CD-A characterization paper lists Taupe as measuring apparent relative dielectric permittivity and classifies it as saturation-related, with limited information for EDZ/fault-zone/water-content interpretation [28, PDF pp. 4-5, Tables 1-2].

TDR methods infer water content because liquid water has a high dielectric permittivity relative to mineral grains and air; the classic Topp et al. relationship is a soil-calibration result, not an Opalinus-specific law [21, pp. 574-577]. An accessible TDR review makes the same modelling caution explicit from the measurement side: TDR estimates water content through bulk permittivity, but porous media contain matrix, air, liquid water, and sometimes bound water, and the measured response can also include electrical-conductivity and dispersion effects [17, pp. 444-446]. The local TD slides use a simple volumetric mixing expression

$$\epsilon_r = \epsilon_{\text{rock}}(1 - \phi) + \epsilon_w \phi S_w, \quad \epsilon_w \approx 80,$$

which is useful as a first-order link to saturation but should not be treated as a fully calibrated Taupe forward model [26, PDF p. 8].

The current workflow therefore writes an explicit Taupe/TDR trend-operator artifact rather than an absolute saturation likelihood. The script `build_taupe_observation_operator.py` reads the 5,088 Taupe rows, the Taupe line-sample mapping, and the OGS mesh porosity field `n_rd`. It writes `taupe_tdr_trend_operator.csv`, `taupe_tdr_series_summary.csv`, `taupe_tdr_observation_operator_summary.json`, and `taupe_tdr_observation_operator.md`. For each of the 24 sensor-EDZ-band series, the first three finite workbook values define a baseline; later observations are stored as raw differences, relative changes, and robust standardized anomalies. Only A3 and A4 are mapped into the current local OGS mesh support, giving 2,544 trend-ready rows; A7 and A8 are retained but flagged as outside the present mesh sampling support. The intended model-side diagnostic is the corresponding baseline-normalized trend in band-averaged $\theta_{\text{model}} = \phi S_\ell$, not the raw Taupe number itself.

The follow-up semantics audit `build_taupe_semantics_audit.py` writes `taupe_tdr_semantics_row_audit.csv`, `taupe_tdr_semantics_series_audit.csv`, `taupe_tdr_semantics_group_summary.csv`, `taupe_tdr_semantics_summary.json`, and `taupe_tdr_semantics.md`. It classifies the 2,544 A3/A4 rows as mapped trend-diagnostic candidates for runs with sampled OGS outputs, while still blocking absolute calibration, and the 2,544 A7/A8 rows as outside the current mesh support. The corresponding state-target table therefore contains 5,088 Taupe/TDR rows, but none is active as an absolute state residual in the current candidate objective.

The run-local diagnostic `evaluate_taupe_tdr_trend_diagnostic.py` closes the next comparison layer without changing the objective. It matches the mapped A3/A4 Taupe rows to sampled OGS output times and forms the same-series standardized residual

$$r_{\text{Taupe,trend}} = \widehat{\Delta\theta}_{\text{model}} - \widehat{\Delta T}_{\text{Taupe}},$$

where hats denote within-series robust standardization rather than a calibrated absolute unit conversion. In the direct reference run and the current local-basis incumbent, the diagnostic compares 1,860 rows across 12 A3/A4 sensor-band series; 684 additional mapped A3/A4 rows are outside the available OGS output time horizon, and all 2,544 A7/A8 rows remain outside the present mesh support. The direct-run standardized trend mean absolute residual is 1.8632. This number is a screening diagnostic for candidate behaviour, not a likelihood term, because Taupe units and a sensor/band uncertainty model remain unconfirmed.

The cross-run audit `build_taupe_candidate_discrimination_audit.py` reuses the same diagnostic across the executed run directories that already contain `ogs_state_samples.csv`. It currently evaluates 74 runs, including 66 runs with the full active combined objective. Across this executed candidate family the Taupe standardized-trend MAE spans only 0.03687076560014302. The best Taupe-only run is `adaptive_combined_001_length_0p050m`, with MAE 1.829884354078035; the best active-objective run is `local_basis_sampler_002_basis_024_det_1_0p0075_s_1p000`, with Taupe MAE 1.863211058631399. This small spread means the present candidate family is a weak discriminator for mapped Taupe trends, even before the unit/calibration and weighting gates are solved. The audit outputs are stored in `taupe_candidate_discrimination_audit.csv`, `taupe_candidate_discrimination_series.csv`, `taupe_candidate_discrimination_summary.json`, and `taupe_candidate_discrimination.md`, with catalogue copies under the Taupe/TDR measurement folder.

The grouped-weight part of the Taupe gate is now tested by `build_taupe_series_weight_sensitivity_audit.py`. The audit uses the series-level cross-run diagnostic for the 12 compared A3/A4 series over the 66 full active-objective

runs. Because every compared series currently contributes 155 rows, aggregate row weighting, equal-series weighting, equal-sensor weighting, and equal-EDZ-band weighting all keep `adaptive_combined_001_length_0p050m` as the best Taupe trend-screen run. The sensor-specific rankings are not identical: A3-only weighting prefers `local_bracketed_003_length_0p031m`, A4-only weighting prefers `adaptive_combined_002_length_0p050m_shift_0p750`, and the 12 individual series have eight distinct best runs. Thus the local evidence does not support a single hard Taupe residual weight yet; sensor-specific uncertainty and grouped weights remain part of the calibration gate. A7/A8 also remain outside the current local mesh support.

The same operator table records candidate absolute conversions for audit only. If the Taupe value is interpreted as volumetric water content in percent, 2,120 of 5,088 rows give mapped-band saturations in the physical interval $[0, 1]$. If the value is interpreted as a Topp-style dielectric permittivity, none of the rows gives physical mapped-band saturation for the current porosity field, which is a warning against using that generic soil calibration as an Opalinus Clay law. The local linear mixing expression with $\epsilon_{\text{rock}} = 6$ gives physical mapped-band saturations for 2,544 rows. These physical-range counts are useful sanity checks but do not identify the correct calibration; a sensor-specific Taupe/ARDP unit statement is still needed before absolute residual weights are assigned.

Model entry should therefore be trend-oriented at first:

- Compare simulated ΔS_ℓ or $\Delta(nS_\ell)$ in EDZ bands with Taupe temporal trends, using `taupe_tdr_trend_operator.csv` as the current row-level operator artifact.
- Use closed-twin stability and open-twin seasonal amplitudes as qualitative or semi-quantitative validation.
- Do not force absolute Taupe values into the likelihood until the ARDP/Taupe workbook columns are mapped to a calibrated dielectric-to-water-content relation.

9.7 Suction and relative humidity: pressure boundary and retention curve

The suction/RH folder contains the open-twin sheets `OT_RH5.xlsx` through `OT_RH8.xlsx`, a suction-location figure, the phase 30 report, TD slides, and minutes. The four RH files cover late 2021/early 2022 to 2025-09-04. RH5 and RH6 stay near 93–96%; RH7 and RH8 include low values near 13%, which should be treated as likely outliers or sensor/logger failures until confirmed. The processed `rh_open_twin_kelvin.csv` table contains 4,247 daily sensor rows, flags low-RH outliers and values above the open-twin 95% reliability caution threshold, and stores the Kelvin-equation liquid-pressure conversion using $T = 298.15 \text{ K}$ and $\rho_\ell = 1095 \text{ kg m}^{-3}$.

The current OGS model already uses a time-dependent pressure curve on the open niche. The active include in `08_curves.xml` is `08_08_open_niche_seasonal.xml`; the shifted `open_niche_seasonal_curve_shifted.xml` is present but commented out in the recovered projection setup. The active curve contains 845 liquid-pressure values ranging approximately from -5.23 MPa to -53.24 MPa . Those values are consistent in scale with a Kelvin-equation conversion from relative humidity to liquid pressure/suction rather than with atmospheric pressure. The Kelvin relationship follows from vapour-pressure equilibrium over a curved liquid surface [20, pp. 448–452]. In Richards form, gas pressure is treated as constant and $p_c = -p_l$; therefore an RH-derived suction can be applied as a liquid pressure boundary $p_l = p_g - p_c$, or as a negative gauge liquid pressure if p_g is absorbed into the convention. Zieffe et al. explicitly describe the CD-A model as using Richards' equation, Van Genuchten retention, and Kelvin-equation boundary conversion for desaturation around the niches [29, pp. 4447–4448].

The working folder now audits this boundary condition explicitly in `inversion_workflow/scripts/audit_rh_boundary_curve.py`. Using the timestep comment $t_0 = 2019-09-18$, the script compares the active OGS curve with the processed OT_RH5–8 Kelvin pressures. Of 4,247 RH rows, 2,280 lie inside the active curve time range, 1,948 later rows are outside it, and 19 low-RH rows are excluded as outliers. Over the overlapping period the median absolute residual between RH-derived gauge liquid pressure and the OGS curve is 13.00 MPa , with a mean absolute residual of 16.32 MPa . Therefore the current open-niche curve should not be claimed as a verified reconstruction of the local RH workbooks; it is an existing model boundary condition whose provenance or preprocessing needs confirmation from BGR/Gesa before it is used as a fixed forcing in a calibrated inversion.

The follow-up semantics audit `build_rh_semantics_audit.py` writes `rh_measurement_semantics_row_audit.csv`, `rh_measurement_semantics_sensor_summary.csv`, `rh_boundary_curve_semantics.csv`, `rh_measurement_semantics_summary.json`, and `rh_measurement_semantics.md`. It records 4,228 valid non-low-outlier RH rows, 19 low-RH outlier rows, and 2,492 open-twin rows above the 95% thermo-hygrometer caution threshold. Inverting the active OGS curve through the same Kelvin coefficient gives an implied RH range of 70.23–96.59%; 772 of the 845 active curve rows are drier than the clean RH5/RH6 workbook minimum. This makes the provenance problem more specific: the active curve may include older, filtered, or different boundary humidity data, but it is not a direct reconstruction of the current OT_RH5–8 sheets.

The request package `build_rh_boundary_provenance_request.py` turns that mismatch into an external information request. It writes `rh_boundary_provenance_request.csv`, `rh_boundary_provenance_evidence.csv`, `rh_boundary_provenance_request_summary.json`, and `rh_boundary_provenance_request.md`, with catalogue copies in the `suction/RH_derived_files` folder. The package has six request rows and ten evidence rows. It asks for the active curve

source data and generation script, time-axis and extension policy, RH sensor screening, Kelvin temperature/density constants, open/closed curve mapping, and the retention-calibration release gate. The generated summary dates the active curve from 2019-09-18 to 2023-12-26 in model time, while the copied RH workbooks continue to 2025-09-04.

The local workflow now also writes candidate RH-derived boundary curves with `build_rh_boundary_candidate_curves.py`. These are not accepted replacement forcings; they are reproducible alternatives that expose what the currently copied sensors would imply. Six sensor-policy curves are generated, with OGS-style XML snippets in `rh_boundary_candidate_curve.xml`. The policy-preferred candidate is `rh5_rh6_median`, because RH5/RH6 are the cleanest copied open-twin thermo-hygrometer streams. It contains 1,063 daily rows from 2021-12-16 to 2025-09-04, 576 overlap rows against the active curve, 487 rows after the active curve ends, and an overlap MAE of 15.15 MPa. This reinforces the current gate: a locally reproducible RH-derived forcing can be prepared, but it should not replace `08_08_open_niche_seasonal.xml` or enter a retention likelihood until the active curve provenance, time axis, sensor screening, conversion constants, and extension policy are confirmed.

The candidate policies are also compared as an uncertainty envelope by `build_rh_boundary_uncertainty_audit.py`. It writes `rh_boundary_uncertainty_envelope.csv`, `rh_boundary_uncertainty_audit.csv`, `rh_boundary_uncertainty_summary.json`, and `rh_boundary_uncertainty.md`, again with catalogue copies in the suction/RH derived_files folder. Across 1,064 daily envelope rows, 577 overlap the active OGS curve and 487 extend beyond it. In the overlap, the local candidate pressure envelope has median width 2.10 MPa and 90th-percentile width 2.20 MPa, while the active curve falls outside that local RH-derived envelope on 575 of 577 dates. The active curve’s mean absolute mismatch to the candidate-envelope median is 15.22 MPa. This is not a reason to tune a residual directly against RH yet; it is a quantitative expression of the provenance gate and supports requesting the original curve-generation data/script before releasing boundary or retention parameters.

For inversion, RH/suction data enter in two ways:

- **Boundary forcing:** reconstruct or audit the open-niche pressure curve from measured RH and temperature. This is not a calibrated parameter unless the RH measurement uncertainty is explicitly included.
- **Retention validation/calibration:** compare measured suction/RH states with OGS $S_\ell(p_c)$ from the Van Genuchten curve. The current XML uses residual liquid saturation $S_{\ell r} = 0.1$, residual gas saturation 0, exponent $m = 0.45$, and $p_b = 10$ MPa, consistent with the van Genuchten form [22, pp. 892–895].

The 2026 slides state that thermo-hygrometers are reliable in the open twin below 95% RH, while psychrometers are preferred in the closed twin above 95% RH. That reliability split should be encoded in the measurement error model [27, PDF p. 6].

9.8 Coordinates, bedding, and structural constraints

The coordinate folder provides the essential measurement-to-model map. The September coordinate workbook has 109 entries: 19 NMR, 8 suction, 8 Taupe, and 72 permeability entries. It contains original/3D coordinates plus two 2D model coordinate blocks. Email context says the `2D_Model (x/y)` columns are the relevant block for the current HERMES modelling. The November TeamBeam coordinate workbook connects NMR, Taupe, and characterization borehole labels to BGR model coordinates. The processed `measurement_coordinates_xy.csv` and `borehole_coordinates.csv` tables keep these coordinate rows in flat form for mesh lookup. The working folder now also contains `inversion_workflow/scripts/build_mesh_observation_lookup.py`, which maps those rows to the projected OGS bulk mesh. It writes point lookups for 107 catalogue coordinate rows, endpoint lookups for 35 NMR/characterization/Taupe labels, and 0.1 m line samples for 8 borehole or Taupe segments. The projection mesh is a local $[-5, 5] \times [-5, 5]$ m domain; in the current lookup, 50 of the 107 catalogue coordinate rows fall outside that mesh bounding box. These outside flags are not errors in the catalogue, but they prove that only a subset of the collected coordinate information can be compared directly with the current Niche-4 OGS domain.

The bedding/geology folder provides the structural prior. Gesa’s email gives an approximate bedding angle of 144° ; the characterization papers describe the upper sandy facies, fault zones, bedding-parallel structures, and the scale dependence of geophysical and permeability measurements. The CD-A characterization paper states that the niches are oriented perpendicular to bedding strike and that the measurement campaign includes ERT, MSM, NMR/CCM, Taupe, pulse tests, suction, and deformation monitoring [28, PDF pp. 2, 4–5]. For inversion this means the anisotropy angle should be bedding-informed, while local anomalies near faults should be either represented as heterogeneous permeability features or assigned larger observation uncertainty.

9.9 Projection workflow and mesh-based parameter fields

The folder `model_projection_inputs/source_files` is not a measurement folder in the physical sense, but it is the most concrete implementation bridge between the measurement catalogue and the frozen OGS process. The projection archive has also been copied into the report working folder at `GESA_model_original/projection_on_mesh_2025-09-05`. Its

`README.txt` states that the project file was changed to call a projected mesh, that `03_parameters_TRM.xml` contains the function/projection specification, and that `bulk_all.vtu` domain identification must be re-done against the projected mesh with `identifySubdomains`.

The script `generate_projections.py` reads `bulk.vtu` with `meshio` and writes `bulk_w_projections.vtu` [26, PDF p. 20]. It creates cell-data arrays called `n_rd` and `k_i_rd` on the triangle elements, using the mesh `MaterialIDs`. The corresponding `03_parameters_TRM.xml` switches `phi` and `k_i` from constants to OGS `MeshElement` parameters with field names `n_rd` and `k_i_rd`; the project file then loads `bulk_w_projections.vtu` as the bulk mesh. This is consistent with the OGS parameter mechanism: official OGS documentation states that `MeshElement` parameters are defined by cell-label data arrays in a loaded mesh, selected through `field_name` [14, Sec. “MeshElement and MeshNode”].

As a reproducible replacement for the random scalar example, the working folder now contains `inversion_workflow/scripts/generate_anisotropic_permeability_field.py`. That script writes a four-component cell-data tensor

$$\mathbf{k_i_rd} = (k_{xx}, k_{xy}, k_{yx}, k_{yy})$$

using the same component order as the constant XML tensor `1.00E-19 0.00 0.00 0.40E-19`. It parameterises the field by a lognormal geometric-mean permeability, a global anisotropy ratio, and a bedding angle. The example configuration uses $\theta = 144^\circ$, $r = 2.5$, and $k_{\text{ref}} = 6.32 \times 10^{-20} \text{ m}^2$, matching the current XML reference values and Gesa’s bedding-angle note.

For the requested inversion this archive should be treated as a prototype, not as the final parameter generator:

- It proves that heterogeneous fields can be introduced without changing the TRM process equations: replace the constant XML parameter by a mesh property with the same physical meaning.
- The recovered random `k_i_rd` field is scalar, isotropic, uncorrelated, and sampled in linear space. The new workflow script replaces it with a log-space, spatially correlated tensor field; an actual inversion driver should still replace the prior/proposal draw by sampled or optimized fields conditioned on the data.
- The script has no fixed random seed, so it is not reproducible as written.
- The porosity-field example is useful only if porosity is later released as an unknown. For the first permeability-focused inversion it should remain fixed unless the data prove permeability alone is not identifiable.
- Before running the projected model, `bulk_all.vtu` and all boundary subdomains must be checked after projection; otherwise boundary conditions can be applied to stale or mismatched submeshes.

The workflow now also prepares an OGS-ready run from the direct permeability diagnostic field in `inversion_workflow/runs/direct_fit_observation_run`. This run copy loads the fitted `bulk_w_projections.vtu` and activates output of `pressure`, `saturation`, `temperature`, `displacement`, and `porosity`, which are the minimum fields needed for later RH/suction, ERT/NMR, Taupe, and HM observation operators. The original GESA project files remain unchanged; the output-variable edit is confined to the run directory and changes only what OGS writes. The run-input audit `inversion_workflow/scripts/audit_ogs_run_inputs.py` confirms that the project mesh files exist, boundary mesh references resolve, the required output variables are requested, and `bulk_w_projections.vtu` contains both `k_i_rd` and `n_rd`. The enhanced audit also checks every VTU XML header for point and cell counts plus declared `DataArray` entries, verifies that the copied boundary/support submeshes expose `bulk_node_ids` and `bulk_element_ids`, and records that the Dockerized-Apptainer OGS run accepted these project inputs with return code 0. It still flags a reproducibility warning: `bulk_all.vtu` and the copied boundary/support VTUs are present but fail local `meshio` decoding because of appended base64 padding, so downstream Python tooling should not depend on `meshio` reads of those submeshes unless they are regenerated with `identifySubdomains`. The local environment does not currently contain an `ogs` host executable: `audit_ogs_environment.py` searched `PATH`, `/usr/bin`, `/usr/local/bin`, `/opt`, and `/home/ber0061` to depth 5. The same audit now records `ogs_container_found_runtime_available` because it finds the collected container image `cda_knowledge_base/file_transfers/collected/2025-04-04_2d_model_container/apptainer_ogs6.5.4.sif`, whose embedded label identifies `ogs/ogs@6.5.4 (cb5b3235101edecf3ba55e1039fe3c19bc13c636)` and `/usr/local/ogs/build/bin` on the container path. No Apptainer/Singularity or `run-singularity` runtime is installed locally, but the preferred backend is now the already used `docker_apptainer_sif` path: Docker runs the collected SIF through `ghcr.io/apptainer/apptainer:latest`. The script `inversion_workflow/scripts/run_ogs_model.py` now records that Dockerized-Apptainer command explicitly, including the workspace bind, numeric user mapping, SIF path, and optional timeout. The initial bounded 60-second startup run with this backend parsed the recovered OGS 6.5.4 project, entered the coupled TRM solve, and wrote seven partial `.vtu` files; it has since been superseded by complete direct, regularized, and first adaptive candidate runs. The post-run sampler `inversion_workflow/scripts/sample_ogs_state_outputs.py` inventories the full OGS outputs and samples 448 lookup rows per output time step: 107 catalogue measurement points plus 341 borehole/Taupe line samples. The sampled outputs contain pressure, saturation, temperature, and displacement fields; porosity is still not available as a direct OGS output field in these VTUs, so the sampler uses the fixed support-mesh `n_rd` porosity fallback for the model-side proxy $\theta = \phi S_l$. This activates 192 NMR rows in complete candidate runs, but θ must still be interpreted with the NMR bound/interlayer-water caveat discussed above.

Operationally, the clean route is to generate one projected mesh per inversion sample or per ensemble member, preserve the original May 2025 XML semantics, and change only the parameter source from a constant to mesh-cell data. Observation operators for ERT, NMR, Taupe/TDR, permeability intervals, and RH/suction should then read simulation outputs at the monthly timesteps documented in the projection archive rather than trying to force all measurements onto the OGS mesh beforehand.

The working folder implements this route in `inversion_workflow/scripts/prepare_ogs_run.py`. The script creates a self-contained run directory, copies the projection model, regenerates `bulk_w_projections.vtu`, and writes `RUN_MANIFEST.json`. In the current smoke test the generated bulk mesh contains `k_i_rd` as a four-component tensor field on all 10,239 `triangle6` cells and `n_rd` as a fixed porosity field, so the projection XML has both mesh fields it reads.

For the non-permeability data, `build_state_observation_targets.py` now writes `state_observation_targets.csv` and `state_observation_samples.csv`. These tables do not force all measurements into the same residual form. They record NMR as point $\theta = \phi S_l$ targets, RH as Kelvin-pressure boundary forcing, Taupe/TDR as baseline-normalized EDZ-band trend targets with absolute calibration pending, and ERT as external resistivity-field comparisons with calibration, spatial lookup, and a direct-run diagnostic ready under explicit transform/support assumptions.

The next workflow layer is now active for NMR where the OGS run covers usable observation times and mapped support. The completed Dockerized-Apptainer runs produce 83 output VTUs and 37,184 sampled state rows per executed candidate. The OGS VTUs contain pressure, saturation, temperature, and displacement but not an explicit porosity field, so `sample_ogs_state_outputs.py` uses the fixed support-mesh cell field `n_rd` from `bulk_w_projections.vtu` as the porosity fallback. The script `evaluate_state_observation_targets.py` writes 11,490 evaluation rows; for the current best lower-support loop candidate, 192 NMR rows are finite objective residuals against $\theta = \phi S_l$, while 4,247 RH rows remain boundary forcing, 1,691 ERT rows remain external diagnostic targets pending transform/support/uncertainty confirmation, 5,265 Taupe/TDR or unsupported rows are not usable for the current state fit, and 95 NMR rows are outside the output time tolerance. `assemble_inversion_objective.py` then combines this state layer with the direct permeability fit. For the best executed lower-support loop candidate, the combined objective has two active components: 75 direct pulse-test rows with effective duplicate-aware weight 52 and objective value 269.84 for $\sigma = 0.5 \log_{10}$ permeability units, plus 192 sampled NMR state rows with objective value 2886.54. The total active objective is 3156.37.

The model-entry coverage audit in `inversion_workflow/measurement_operator_coverage.md` makes this status explicit across all collected measurement streams. It lists nine observation groups. For the current best executed lower-support loop candidate, two groups have active objective rows: the direct permeability pulse tests and sampled NMR state targets. ERT is inventoried and matched to field files and has theta-to-resistivity calibration, a spatial lookup, and direct-run plus 66-run cross-candidate log-resistivity diagnostics, but still needs transform/support/uncertainty confirmation before assigning residual weights; Taupe/TDR is mapped to EDZ-band line samples and has a baseline-normalized trend operator with a run-local A3/A4 comparison diagnostic, but remains outside the active objective until the absolute unit/calibration is confirmed; RH has been converted to Kelvin pressure, audited as boundary forcing, and expanded into local candidate boundary curves, but it remains outside the active objective until the active-curve provenance and extension policy are confirmed. Coordinates, bedding/geology, and the model-projection inputs are support layers. The remaining HM monitoring streams now have a structured layout/qualitative layer and levelling summary rows, but they remain outside the active objective until the Geoscope time-series exports and laser-scan statistical files are located.

The generated model-entry matrix in `inversion_workflow/measurement_model_entry_matrix.md` joins this coverage map with the residual/likelihood semantics, stream activation gates, and final include/exclude register. It keeps the nine collected measurement classes on one row each: two classes currently contribute 267 active objective rows, three classes are diagnostic or boundary-audit only, three are support/prior layers, and other HM monitoring is not ready for hard residual weights. The matrix is therefore the report-facing index for deciding whether a measurement is active, diagnostic, support-only, boundary-only, or awaiting a collaborator/provenance decision.

The separate likelihood/activation audit in `inversion_workflow/measurement_likelihood_model.md` records the stricter statistical decisions that should not be inferred from data coverage alone. It has seven stream-level rows and two active streams: direct permeability pulse tests and sampled NMR, for 267 active objective rows. The active permeability residual is $\log_{10} k_{\text{pred}} - \log_{10} k_{\text{obs}}$ with the current broad $\sigma = 0.5 \log_{10}$ scale and duplicate-aware weights. The active NMR residual is still caveated: the bound-water sensitivity audit rules out a naive absolute $\theta = n S_\ell$ residual, and the 66-run bias/anomaly audit shows that a bias-safe NMR diagnostic changes the top candidate. NMR therefore requires either a trend/anomaly residual or explicit bound/interlayer-water bias or model-error term; ERT and Taupe/TDR are kept as forward observation operators until support/calibration choices are confirmed; RH remains a boundary-condition provenance audit rather than a cell-wise residual.

A dedicated stream gate audit now makes these decisions machine-checkable in `inversion_workflow/measurement_stream_activation_gate_audit.md`. It records eight stream/support layers, 20 gate checks, and seven required failed gates. The only streams with active objective rows are direct permeability and NMR, both with tracked caveats. ERT and Taupe/TDR are diagnostic-only, RH is boundary-audit-only, and other HM monitoring is not ready for hard residuals because no Geoscope, laser-scan, or full levelling numeric exports with epochs, units, support, and uncertainty are present. This audit is the current blocker list for promoting the diagnostic streams into a weighted all-measurement objective.

The blocker list has been converted into a reproducible closure package in `inversion_workflow/measurement_gat`

`e_closure_request.md`, with the machine table `inversion_workflow/measurement_gate_closure_request.csv` and an email-ready draft in `inversion_workflow/measurement_gate_closure_email_draft.md`. It contains 13 closure items: seven external BGR/provider requests and six internal modelling decisions. The high-priority requests cover the ERT coordinate transform/support mask and covariance model, Taupe workbook unit/calibration, RH active-curve provenance and uncertainty policy, Geoscope/laser/levelling numeric exports and metadata, and the final NMR bound/interlayer-water residual definition. The internal decision register also records the separate NMR not-promoted-default policy for the current report state. The medium-priority items record the permeability gas-pulse model-error policy, optional historical permeability endpoint geometry, and Taupe grouped weighting/support decisions. Until these items are closed, the report should describe the current fit as a direct-permeability plus conditional-NMR inversion with ERT, Taupe/TDR, RH/suction, and other-HM streams used as diagnostics or provenance checks.

Before treating those rows as external-only, the same gate criteria were applied to the local source catalogue and to raw `/home/ber0061/Downloads` files. The catalogue recovery audit in `inversion_workflow/local_gate_recovery_audit.md` rescanned 2,540 source/index documents and found zero possible gate-closing evidence rows. The Downloads recovery audit in `inversion_workflow/download_gate_recovery_audit.md` scanned 533 CD-A/HERMES-relevant filename, archive-member, Office-XML, and text documents and also found zero possible gate-closing evidence rows. Its companion inventory `inversion_workflow/download_gate_recovery_inventory.csv` records 25 SHA1-verified duplicate/catalogue rows, including `ERT_meas_Niche_open(1).zip`, `ERT_meas_Niche_open(2).zip`, and `003_Nov_2025(1).zip`, plus the extracted/run-output directory `/home/ber0061/Downloads/CDA_N4_2D_250403`. That directory contains run-output VTUs and duplicated base model files rather than provider answers to the transform, uncertainty, calibration, provenance, numeric-export, or endpoint-geometry gates. Thus no currently known local download changes the activation status of the seven external gates.

The seven external rows are now split into five recipient-specific drafts in `inversion_workflow/external_gate_request_pack.md`, with suggested contact routes resolved from the local mail index. All five drafts have a suggested To route through `Gesa.Zieffle@bgr.de`; the ERT draft additionally lists `Markus.Furche@bgr.de` as suggested Cc, while the remaining provider routes keep explicit forwarding caveats because no more specific sender was resolved locally. The response-intake tracker in `inversion_workflow/external_gate_response_intake.md` creates stream-local provider-response folders and records one response-note template, acceptance test, and refresh-command list per external request. The dispatch-readiness audit in `inversion_workflow/external_gate_dispatch_audit.md` checks those drafts against the intake tracker before sending: all seven requests are ready across the five drafts, all seven rows carry suggested To routes, zero dispatch checks fail, and the Gmail draft tracker `inversion_workflow/external_gate_gmail_drafts.csv` records five Gmail drafts covering all seven request rows. No request is recorded as sent and all seven responses remain missing. A point-in-time Gmail live-state audit in `inversion_workflow/gmail_gate_live_state_audit.md` then checked these same threads plus the separate CTE confirmation draft. At 2026-06-01 03:55:46 CEST, all six expected Gmail drafts were still present as drafts; a sent-subject search found zero sent copies; the recent named-provider search found zero replies from the Gesa, Markus, or Tuanny routes; and the recent CD-A/HERMES/TeamBeam context search returned only the generated drafts. This confirms the current response-intake status but does not replace sending or answering the requests.

The consolidated send/response state is now captured in `inversion_workflow/external_blocker_dashboard.md`. It joins the external request, response-intake, dispatch, Gmail live-state, local-recovery, Downloads recovery, and CTE confirmation trackers into eight open blocker rows: seven external measurement gates and one CTE confirmation row. All eight rows remain unsent and missing-response blockers; the dashboard therefore provides the current worklist, not closure evidence.

The current executed-candidate evidence is also joined in `inversion_workflow/cross_stream_candidate_scorecard.md`. The scorecard compares 66 common run ids across the active objective, the NMR bias/anomaly diagnostic, ERT log-resistivity MAE, and Taupe/TDR trend MAE without changing the likelihood. It finds no candidate in the top 10 of every stream. The active incumbent `local_basis_sampler_002_basis_024_det_1_0p0075_s_1p000` ranks first in the active objective but only 14th in the NMR bias/anomaly diagnostic, 8th in ERT, and 29th in Taupe/TDR. The best mean-rank compromise is `local_basis_sampler_003_basis_029_det_1_0p0100_s_1p000`, with mean rank 10.4 and worst rank 18. This is the present evidence that the active-objective incumbent should remain conditional on the unresolved diagnostic gates rather than being reported as the final all-measurement permeability field.

The same pause point is now checked by `inversion_workflow/runs/cross_stream_hybrid_field_plan/CROSS_STREAM_HYBRID_FIELD_PLAN.md`. That screen builds 15 log-magnitude hybrids from five cross-stream winner runs and the active incumbent. It derives magnitude from the actual `k_i_rd` tensor, not from the stale `k_mag_rd` metadata field, and then preserves the active tensor orientation and anisotropy ratio by scalar multiplication. The best hybrid, `cross_hybrid_mean_rank_all_streams_plus2_a0p25`, has direct objective 269.8180571059851, i.e. zero improvement over the active direct objective at the reported precision. Thus the cross-stream hybrids are useful as deliberate diagnostic probes, but they do not justify another OGS batch for the active direct-permeability plus raw-NMR objective alone.

The parameter-release audit in `inversion_workflow/inversion_parameter_release_plan.md` then connects those measurement decisions back to the actual OGS XML. It writes 14 rows from the base `03_parameters_TRM.xml`, media files, projection `MeshElement` parameter file, and selected regularized candidate run. The result is deliberately restrictive: the only stage-1 active unknown is the `k_i_rd` tensor-magnitude field; the `n_rd` porosity field is a fixed support field even though sampled NMR state residuals are active; tensor shape, van Genuchten p_b , and exponent stay gated until

the final NMR policy, support decisions, and RH boundary-curve provenance are settled; elasticity, Biot, and swelling remain later mechanical validation candidates; and the open-niche pressure curve plus the suspicious CTE value are confirmation/provenance blockers, not ordinary calibration parameters. The same audit now records two active objective components with objective 3156.353066948979, including 75 direct permeability rows and 192 sampled NMR rows from 83 OGS output times, and it blocks further same-support active-objective OGS batches until support, likelihood, bounds, or stream-gate evidence changes. The dedicated `inversion_workflow/cte_confirmation_request.md` package now turns the CTE blocker into a provider-facing question routed to Gesa with Tuanny as the model-setup contact in copy. It asks whether CTE=1254.74 is an intended thermal-expansivity value in $1/K$, a copied `c_p_s` heat-capacity value, inactive in the intended run, or another unit/convention. Until the response is recorded, the frozen XML value is kept unchanged and uninterpreted.

The same staging decision is now enforced on prepared runs by `inversion_workflow/scripts/audit_inversion_release_gates.py`. The audit is called inside the candidate driver and also writes a global summary in `inversion_workflow/inversion_release_gate_audit.md`. For the current three regularized candidate directories, first three adaptive candidate directories, two local-refinement candidate directories, three local-bracketing candidate directories, three optimizer-proposed candidate directories, three continuous-proposed candidate directories, three lower-support continuous candidate directories, six lower-support loop candidate directories, six broad continuous candidate directories, three local-basis candidate directories, and thirty production sampler candidate directories it passes 1300 checks: `k_i` is the active `MeshElement` field `k_i_rd`, `phi` remains fixed support `n_rd=0.105`, media and retention XML files match the projection reference, and no blocked or later-release parameter has been silently changed. The consolidated guardrail audit in `inversion_workflow/frozen_model_measurement_inclusion_audit.md` then links that release control back to the measurement catalogue. It checks 75 `RUN_MANIFEST.json` files, the 65 release-gated OGS candidate runs, the current positive-definite permeability-field package, the 206-file measurement-info mirror, 1,880 indexed archive members, and 34 workbook-sheet summaries. The audit passes, but it also records that the current field remains an active-objective incumbent rather than a final all-measurement inversion because the ERT, Taupe/TDR, RH, other-HM, endpoint-geometry, and CTE gates remain open.

Finally, the full objective is tracked by `inversion_workflow/objective_readiness_audit.md`, generated with `build_objective_readiness_audit.py`. This audit is intentionally stricter than a progress note: it maps the original report/model/inversion request to 13 evidence rows. At the present state it reports `completion_state=not_complete`. The clean report build, source-model recovery, measurement inventory, processed observation tables, source coverage, release plan, and release gate are in place. The recovered OGS 6.5.4 SIF has now completed direct, regularized, first adaptive, local-refinement, local-bracketing, and optimizer-proposed candidate runs through continuous, lower-support continuous, repeatable lower-support loop, and broad continuous candidate runs through Dockerized Apptainer, and sampled NMR rows are active in the combined objective. The full inversion remains incomplete because the current comparison is still proposal/sampler-batch selection rather than a converged posterior or optimizer trace, ERT/Taupe/RH remain gated, and the diagnostic streams have not been calibrated into a common likelihood. The remaining report and modelling questions are also consolidated in `inversion_workflow/open_question_resolution_matrix.md`, generated with `build_open_question_resolution_matrix.py`. That matrix links the original open-question list, source/citation checks, report-comment audit, external gate trackers, and final-promotion criteria. It records 23 rows: 6 locally resolved/current-scope rows, 9 external send/response rows, 6 internal policy/final-decision rows, one tracked current-policy caveat, one deferred time-dependency row, and an explicit no-new-evidence closeout decision row. It therefore turns the remaining questions into provider-response, modelling-policy, or conservative-closeout gates rather than unresolved report comments. The local-basis sampler is now included as executed combined evidence: its selected three-candidate batch passed release gates and shifted the active best combined objective to 3156.35307. The current field is packaged separately in `inversion_workflow/current_permeability_field/CURRENT_PERMEABILITY_FIELD.md`, with the copied `bulk_w_projections.vtu` mesh, summary JSON files, cell-wise statistics for the actual `k_i_rd` tensor, detailed direct and sampled NMR residual tables, and a run-local `ogs_run_inputs/` snapshot of the exact project file, XML includes, meshes, and curves used for the executed incumbent. The accompanying `packaged_file_manifest.csv` records SHA256 checksums for the packaged field, audits, residual tables, summaries, and OGS input snapshot. The generated `CURRENT_FIELD_REPRODUCIBILITY_AUDIT.md` verifies 19 package, manifest, snapshot, objective, release-gate, execution, and field checks with zero required failures. The package records 10,239 positive-definite `triangle6` tensor cells, fixed anisotropy ratio 2.5, bedding angle 144° , fixed support porosity $n = 0.105$, and the same active objective value. It is therefore the present deliverable candidate field for inspection and reruns, but still not a final all-measurement inversion field. The visual inspection artifact in `inversion_workflow/current_permeability_field/visual_inspection/CURRENT_FIELD_VISUAL_INSPECTION.md` renders six cell-wise PNG maps from that packaged VTU, including log10 geometric-mean permeability, minor and major eigen-permeability, local-basis increment, nearest-anchor distance, and local-basis weight support. These maps are used as QA for the active-objective field and do not change the promotion status. The promotion decision is made machine-checkable in `inversion_workflow/current_field_selection_audit.md`: that audit accepts the packaged field as the current active-objective incumbent, but refuses promotion to a final all-measurement field because the provisional NMR trend/anomaly residual and the cross-stream mean-rank scorecard prefer different runs, while ERT, Taupe/TDR, RH, other-HM, and external-response criteria remain gated. The companion scenario audit `inversion_workflow/conditional_field_selection_scenarios.md` enumerates eight gate/outcome scenarios; the packaged field wins only the current raw-NMR active objective, while five different fields win across the scenarios. This makes the non-final status a consequence of the measured stream

conflicts, not just a wording caveat. Those five scenario-winning meshes are copied with common tensor statistics in `inversion_workflow/conditional_field_candidates/CONDITIONAL_FIELD_CANDIDATES.md` so the current active-objective field, promoted-NMR field, ERT-screen field, Taupe-screen field, and mixed diagnostic field alternatives can be inspected without modifying the exchanged GESA model. The same package now writes per-candidate `diagnostic_metric_evidence.csv` files and a combined `conditional_field_candidate_metric_evidence.csv`: 25 active-objective, NMR-label-bias, NMR-trend/anomaly, ERT, and Taupe metric rows are extracted from the global diagnostic audits, with no missing metric-evidence rows for the five scenario winners. The follow-up field-difference audit in `inversion_workflow/conditional_field_candidates/CONDITIONAL_FIELD_DIFFERENCE_AUDIT.md` compares the four non-current scenario winners against the current field cell by cell using the geometric-mean $\log_{10}$ permeability of $k_{i,rd}$. One alternative, `local_basis_sampler_003_basis_029_det_1_0p0100_s_1p000`, is essentially the same local-basis field at this scale: its mean absolute difference is 8.5×10^{-6} log units and no cell differs by more than 0.05 log units. The broader gate-dependent alternatives are local but material: the largest mean absolute difference is 0.0377448 log units, with at most 241 of 10,239 cells exceeding 0.05 log units and 233 cells exceeding 0.10 log units. This supports keeping the current field available for inspection while still refusing to label it as an all-measurement inversion result. The final-promotion gate is consolidated in `inversion_workflow/final_inversion_promotion_checklist.md`. That checklist records 17 criteria: five pass, one passes with the tracked source-model caveat, and two pass only for the active objective; six remain externally blocked, one remains an internal NMR-residual policy decision, and two fail promotion because the scenario winner is unstable and the current field-selection audit refuses final promotion. Its current decision is therefore `do_not_promote_current_field`. The checklist is the practical close-out map: either the ERT, Taupe/TDR, RH, other-HM, permeability-endpoint, CTE, and NMR-policy gates must be closed, or the modelling team must explicitly record which gated streams are excluded from the final objective. The companion close-out playbook, `inversion_workflow/final_inversion_closeout_playbook.md`, expands the nine open promotion criteria into actions: six existing Gmail draft/response routes, one internal NMR policy decision, and two scenario/final-field decision steps. It lists the response-note locations, acceptance evidence, and regeneration commands needed after replies or explicit exclusions are recorded. The regenerated playbook also threads the direct-permeability support-conflict evidence through the permeability endpoint, scenario-stability, and final-field approval rows, so resolving or explicitly retaining the current support/likelihood policy is a named close-out action before same-support OGS spending or final promotion can resume. The additional decision register `inversion_workflow/final_objective_decision_register.md` records the explicit include/exclude layer for those same open rows. It currently contains nine pending or not-ready decisions: six external stream/provenance decisions, one internal NMR policy decision, and two scenario/final-field decisions. Each row states the current allowed use of the evidence, the include path, the diagnostic-only or exclusion path, the acceptance evidence, the report wording requirement, and the refresh commands. It now also carries the direct-permeability support-conflict numbers and the still-unapproved support/likelihood policy into the endpoint, scenario-stability, and final-field decision rows. Consequently, a gated stream is not silently waived: it must be closed, kept diagnostic-only, explicitly excluded, or waived by a documented modelling decision before the current permeability field can be promoted. The companion final-objective scenario matrix `inversion_workflow/final_objective_scenario_matrix.md` maps those include/exclude choices to nine explicit selection options. The current packaged field wins only the narrow raw-NMR active-objective option and the direct-only permeability tie case; every currently scored option that promotes NMR trend/anomaly or accepts ERT or Taupe/TDR chooses a different run, while RH, other-HM, and endpoint-missing historical permeability remain unscored future options until their provider evidence is filed. The matrix also records that the direct-only tie is not a clean final objective: the current support map has repeated-row conflicts, zero same-support reducible gap, and no approved direct-permeability policy change. The recommendation packet `inversion_workflow/final_objective_include_exclude_recommendations.md` records the conservative no new evidence interpretation of the same rows. It recommends keeping ERT, Taupe/TDR, RH/suction, other-HM monitoring, endpoint-missing historical permeability rows, and the suspicious CTE value out of final hard likelihood weights unless provider evidence arrives or an explicit modelling decision is recorded; it also keeps the current direct-permeability support/likelihood policy as a signoff issue rather than a silent final assumption. This packet is not itself a decision and does not unblock promotion; it keeps the current field labelled as an active-objective incumbent. The further no-new-evidence closeout draft `inversion_workflow/final_objective_no_new_evidence_closeout_draft.md` translates that conservative position into exact decision text for review. It contains nine draft rows and a single-block acceptance text that would choose the narrow `F01_current_raw_nmr_exclude_gated_streams` option only after user/modelling-team approval and regenerated scenario, current-field, and promotion audits. The draft records no actual decisions, sends no mail, and does not promote the current field. The matching acceptance-record template `inversion_workflow/final_objective_no_new_evidence_acceptance_record_template.md` is the signoff guardrail for that path. It has nine fillable rows, zero approvals recorded, and a not-ready-to-apply status, so it prevents the draft text from being mistaken for an accepted final objective. The regenerated draft and acceptance template also carry the direct-permeability support-conflict evidence into the final signoff layer itself: 75 active pulse-test rows occupy 30 OGS support cells, 24 of those cells contain repeated rows, and 16 repeated cells span at least two $\log_{10}$ permeability units. The dominant signoff example is still cell 4648 in BCD-A32 at 0.85–0.87 m, where three active rows span 6.948847 $\log_{10}$ units and include the configured-scalar envelope conflict. Because the permeability likelihood-policy acceptance template still has 0/1 primary approvals and the next field-fit gate marks same-support active-objective batches as non-executable, even the conservative no-new-evidence path cannot be read as approval to rerun the same support map or to promote the active field. For user review before any send action, `inversion_workflow/gmail_draft_send_review_packet.md` consolidates those six Gmail drafts into one table: five external measurement-gate drafts and one CTE confirmation draft, eight covered request

rows, all six marked ready for review, and all six still recorded as unsent. The production sampler/convergence audit now adds five executed production rounds and recommends pausing the active direct-permeability/raw-NMR sampler. The cross-stream scorecard then shows why this pause is not a final inversion claim: the active incumbent does not dominate the NMR bias/anomaly, ERT, and Taupe/TDR diagnostics viewed together. Two cheap follow-up field-family screens were therefore run before spending more OGS time. The cross-stream hybrid screen in `inversion_workflow/runs/cross_stream_hybrid_field_plan/CROSS_STREAM_HYBRID_FIELD_PLAN.md` blends the active field magnitude with diagnostic-winner magnitude patterns and finds no direct-permeability improvement: the best hybrid ties the active direct objective at 269.818057. The structural/EDZ screen in `inversion_workflow/runs/structural_edz_field_family_plan/STRUCTURAL_EDZ_FIELD_FAMILY_PLAN.md` then tests 234 central EDZ caps, central shells, bedding-parallel bands at 144°, and broad BCD-A32/BCD-A33 open-twin corridors. It catalogues 30 active permeability objective support cells and 1,928 ERT ready-support cells, but again finds zero improving direct candidates; the best bedding-band row only ties the active direct objective. This makes the pause stronger: more OGS runs in the current magnitude-field neighborhood are not justified by the active direct screen alone, while ERT, Taupe/TDR, RH, and other-HM gates remain unresolved. The residual conflict audit in `inversion_workflow/permeability_residual_conflict_audit.md` explains why this is not merely a sampler-convergence statement. For the 75 active pulse-test rows it finds weighted RMSE 1.610715 log units, 48 rows with absolute residual at least one log unit, 21 rows at least two log units, and two high-permeability rows above the configured scalar tensor range. It also finds 24 support cells with repeated active rows and 16 cells where repeated observations span at least one log unit. Thus the remaining direct mismatch is a measurement-support, gas-to-water interpretation, uncertainty, or parameterization problem before it is another routine OGS batch. The spatial support-conflict audit in `inversion_workflow/permeability_support_conflict_spatial_audit.md` maps those support cells back onto the current `triangle6` mesh. It confirms that the 75 active rows occupy 30 support cells on the 10,239-cell current field, that 24 of the support cells have repeated rows, and that 16 repeated cells span at least two $\log_{10}$ units. The dominant mapped conflict is cell 4648 in BCD-A32 at 0.85–0.87 m, where three active rows span 6.948847 log units and include the only configured-scalar envelope conflict. This turns the stop rule into a spatially inspectable statement: under the present one-value-per-support-cell map, the largest direct residuals are co-located contradictory observations, not missing smooth-field degrees of freedom. The companion likelihood-policy audit in `inversion_workflow/permeability_likelihood_policy_audit.md` makes this distinction sharper without changing the active objective. Under the recorded duplicate-weighted Gaussian row policy the direct objective is 269.818057 and the top ten rows contribute 0.494 of the row loss. A support-cell weighted-mean diagnostic, however, collapses the 75 active rows to 30 model-support groups and has an objective-like value of 1.77×10^{-28} , while a support-cell median diagnostic remains 134.727790. The current field is therefore matching the duplicate-weighted means of conflicting rows that occupy the same support cells; switching to robust tails, support-cell aggregation, or scalar-range outlier handling must be an explicit likelihood decision rather than an implicit field-fit success. The support lower-bound audit in `inversion_workflow/permeability_support_lower_bound_audit.md` turns this into a field-search limit. Holding the current observation-to-support-cell map fixed, the best possible single log-permeability prediction for each support cell is the duplicate-weighted observed mean. The accepted field already attains that lower bound: the current row-Gaussian direct objective and the single-support lower-bound objective are both 269.818057, with zero same-support reducible gap. All 30 support groups are at their lower bound, and 16 groups still span at least two $\log_{10}$ units. Thus another field that preserves the same support map cannot reduce the dominant row-Gaussian pulse-test loss; reducing it requires changing support, gas-pulse interpretation, likelihood semantics, scalar bounds, or tensor-shape freedom. The decision request in `inversion_workflow/permeability_likelihood_decision_request.md` keeps the rowwise Gaussian policy as the current-report default, but records the required alternatives before further active-objective OGS spending: robust row residuals, support-cell aggregation, explicit scalar-range outlier handling, or a genuinely new field-family gate with a materiality threshold. The configured-scalar outlier disposition in `inversion_workflow/permeability_configured_scalar_outlier_disposition.md` then makes the scalar-bound issue local and explicit. The two active rows outside the configured scalar envelope are one duplicated BCD-A32 0.87 m high-permeability value. It is only 0.107 $\log_{10}$ units above the configured upper envelope, whereas the same OGS support cell spans 6.949 $\log_{10}$ units because a nearby active low-permeability row is mapped to the same cell. The local disposition is therefore not to widen eigenvalue bounds or release tensor shape from these rows alone; they remain visible under the current rowwise Gaussian default until the modelling team accepts that disposition or chooses a robust/capped/support-cell/outlier policy explicitly. The no-OGS existing-field rerank in `inversion_workflow/permeability_likelihood_scenario_rerank.md` then scores 522 materialized permeability-field VTUs under those diagnostic policies. It finds 40 active row-Gaussian best ties, with the current accepted field included in that tie set. Huber and Student-t row policies keep their representative winners inside the tie set, whereas capped row loss, support-cell median aggregation, and configured-scalar inside-only scoring select winners outside it. Thus the rerank is useful decision evidence, but not an approval to change the current-report likelihood policy. The winner cross-stream audit in `inversion_workflow/permeability_likelihood_winner_cross_stream_audit.md` then checks whether those policy winners have executed-run state and diagnostic evidence. Four of seven policy-winner rows have cross-stream scorecard evidence, representing two unique fields. All three outside-tie non-default winners are direct-only fields without OGS/state/ERT/Taupe/NMR scorecard evidence. The row-Gaussian representative selected by candidate-id sorting has active-objective rank 45 and mean all-stream rank 30.4, whereas the current accepted field has active-objective rank 1 and mean all-stream rank 13.2. A direct-only non-default winner therefore cannot replace the current field without OGS execution and stream diagnostics. The next field-fit gate in `inversion_workflow/permeability_next_field_fit_gate.md` makes this stop rule explicit. It records eight gates and recommends pausing same-support active-objective OGS batches under the current rowwise Gaussian policy. Further OGS spending should be reopened only

after a support mapping, likelihood, configured bound/tensor-shape, or accepted measurement-stream objective changes; a non-default direct-policy winner also needs OGS state and stream diagnostics before it can be used for final promotion. The consolidated packet `inversion_workflow/permeability_likelihood_support_recommendations.md` collects the same conclusion into eight direct-permeability decision-support rows. It keeps `keep_rowwise_gaussian_default` as the current-report policy, records zero same-support reducible gap, rejects immediate bound widening or tensor-shape release, and states that the packet does not unblock final promotion. The matching policy acceptance-record template `inversion_workflow/permeability_likelihood_policy_acceptance_record_template.md` is the signoff guardrail for any direct-permeability likelihood/support/outlier policy change. It contains the five exclusive policy options, currently records zero of one required primary approvals, is not ready to apply, records no actual decision, changes no active objective, and does not reopen same-support OGS spending.

9.10 Other HM monitoring

The other HM monitoring folder contains minutes, TD modelling slides, levelling slides, the HERMES input note, and the CD-A Tecplot layout. These sources identify extensometers, mini-piezometers, crackmeters, laser scans, levelling/miniprismas, temperature/RH, and opening-time records. The 2024 HERMES input note states that BGR has long-term measurements of deformation, humidity, water content, pore water pressures, crack width, pointwise piezometer/extensometer data, and two-dimensional ERT/TDR/laser-scan data [19, PDF p. 5].

The working folder now makes this secondary HM layer explicit with `build_other_hm_monitoring_inventory.py`. The script streams `VisualisationCDA.dat` and writes `other_hm_visualisation_zones.csv`, `other_hm_visualisation_text_labels.csv`, `other_hm_levelling_displacements.csv`, `other_hm_qualitative_targets.csv`, and `other_hm_monitoring.md`. The Tecplot layout contains 84 zones and 11 legend labels. The classified zones include 5 extensometer zones, 1 mini-piezometer zone, 22 fracture/crack-geometry zones, 12 miniprisma/geodetic zones, 2 laser-scan surface zones, 6 RH/suction-support zones, and support zones for NMR, Taupe/TDR, permeability, convergence points, evapometer, and niche geometry.

The 2026 TD minutes are important because they distinguish available streams from usable residuals. They state that a Geoscope update sent on 2026-04-20 covered RH, temperature, door openings, extensometer, mini-piezometer, crackmeter, suction, and laser scans with statistical interpretation [27, PDF p. 3]. Those raw Geoscope and laser-scan exports are not present in the current catalogue. The request package `inversion_workflow/processed_observation_s/other_hm_missing_numeric_request.md` therefore lists the missing numeric exports explicitly: mini-piezometer time series for BCD-A28 to BCD-A31, extensometer time series with BCD-A9/B10 failure metadata, crackmeter time series, auxiliary Geoscope RH/temperature/door/suction logs if distinct from the already catalogued RH files, the 2026-04-20 laser-scan statistical interpretation, and a full precision-levelling survey table. These rows remain at zero active objective weight until the numeric files include units, support geometry, reference conventions and quality/status flags. The local absence claim is now checked by `build_other_hm_numeric_source_audit.py`. It scans the ten current other-HM source files, the seven PDF members inside the 2026 presentation ZIP, extracted text, the Tecplot layout support, and the levelling table. All six missing-export request classes have some local support geometry or extracted labels, but zero are hard-residual-ready. The ZIP contributes no spreadsheet/CSV-style numeric members; the only numeric-extension source in this folder is `VisualisationCDA.dat`, which supplies support geometry rather than dated Geoscope time series. The audit therefore keeps mini-piezometer, extensometer, crackmeter, laser-scan, auxiliary boundary-context, and full levelling residuals inactive until the requested exports with units, timestamps or survey epochs, reference conventions, uncertainty, and quality flags are delivered. The same minutes report that mini-piezometers BCD-A28 to BCD-A31 were working well, that BCD-A9/B10 closed-niche horizontal/vertical extensometers failed since September 2025, and that crackmeter data show one ongoing closed-niche trend plus open-niche seasonality [27, PDF p. 3]. They also record the higher-level trend split: the open twin shows ongoing trend in extensometer, suction and Taupe, whereas the closed twin is mostly static in extensometer, pore-pressure and suction measurements except for one crackmeter position with ongoing closure [27, PDF p. 7].

The levelling slides provide the only pointwise numeric deformation summary currently available in this secondary-HM bundle. The extracted table has 12 height-change values from the first 2022-08-29/30 measurement to the 2026-03 campaign. The largest reported changes are CDA-O1 -2.1 mm settlement and CDA-C4 $+1.3$ mm heave/uplift; the reported detectable displacement is $0.1\text{--}0.2$ mm at 95% confidence, depending on the point [18, PDF pp. 5–6]. These values can become deformation-validation targets only after the survey reference frame and the OGS displacement-output support are aligned.

The BGR 2026 modelling slides set the same priority for future HM observation operators: compare with extensometer displacements/strains, mini-piezometer pore pressures, and suction-derived pressure in the desaturated zone [3, PDF p. 2]. They also warn that a simplified perfect-ring 2D model already showed simulated strain discrepancies relative to the extensometer measurements [3, PDF p. 15]. This is a useful guard against accepting a hydraulic/permeability fit that is mechanically implausible.

For the current permeability-focused inversion these streams are secondary, but they should be kept as validation gates:

- Mini-piezometers validate pore-pressure trends after the Geoscope pressure series are located.

- Extensometers/laser scans validate displacement and strain fields, with the September 2025 closed-niche extensometer failure kept as a data-quality boundary.
- Crackmeters and convergence observations constrain whether simulated deformation trends are qualitatively plausible, especially the open/closed twin contrast.
- Levelling values can check the sign and millimetre-scale magnitude of vertical displacement after coordinate/reference-frame alignment.
- Temperature/RH/opening-time records explain boundary-condition disturbances.

9.11 Recommended staged inversion workflow

1. Freeze the May 2025 OGS XML and April mesh/project structure as the base model.
2. Validate the local data inventory with `inversion_workflow/scripts/validate_observation_manifest.py`; this guards against silently fitting against missing or structurally changed measurement files.
3. Regenerate the normalized observation tables with `inversion_workflow/scripts/build_processed_observations.py` after any catalogue update, and use `processed_observations/README.md` as the table index.
4. Build a geometry map from `coordinates_geometry_layout/source_files` or from the processed coordinate CSVs, then run `inversion_workflow/scripts/build_mesh_observation_lookup.py` to separate in-domain measurements from nearest-cell fallback or out-of-domain points.
5. Build explicit non-permeability state targets with `inversion_workflow/scripts/build_state_observation_targets.py`. The current layer contains 11,490 target rows: mapped NMR water-content targets, RH Kelvin-pressure boundary-forcing rows, Taupe/TDR baseline-normalized trend targets, and ERT external field-comparison rows.
6. Quantify the NMR bound/interlayer-water caveat with `inversion_workflow/scripts/build_nmr_bound_water_sensitivity.py` before any absolute NMR water-content row is allowed into the numerical objective.
7. Rebuild the likelihood/activation audit with `inversion_workflow/scripts/build_measurement_likelihood_model_1.py` so each measurement stream has an explicit residual transform, scale/weighting rule, bias/model-error caveat, and activation gate before it can enter the objective.
8. Rebuild the XML-linked parameter-release plan with `inversion_workflow/scripts/build_inversion_parameter_release_plan.py`. This keeps the inverse problem staged: active `k_i_rd` magnitude fields against the direct permeability and sampled NMR objective first, fixed `n_rd` support, no further same-support OGS spending without a gate change, and delayed release of tensor shape, porosity, retention, boundary, or mechanical parameters until the recorded gates are met.
9. Audit prepared run directories with `inversion_workflow/scripts/audit_inversion_release_gates.py`, or use the same gate through `evaluate_inversion_candidate.py`. A failed release gate should stop candidate ranking because the run no longer represents the documented permeability-only stage.
10. Refresh the objective-level readiness audit with `inversion_workflow/scripts/build_objective_readiness_audit.py`. This audit is the current completion check: it should remain `not_complete` until OGS outputs exist, state likelihoods are activated where justified, the five current external-gate Gmail drafts and the separate CTE Gmail draft are reviewed/sent and answered, and the combined inversion has been run.
11. Re-run the local and Downloads gate-recovery audits with `inversion_workflow/scripts/build_local_gate_recovery_audit.py` and `inversion_workflow/scripts/build_download_gate_recovery_audit.py` whenever new TeamBeam, Gmail, Thunderbird, or raw download files are added. A duplicate archive or OGS run-output folder is not a gate closure unless it contains the gate-specific transform, uncertainty, calibration, provenance, numeric-export, or endpoint-geometry evidence and is filed through the response-intake tracker.
12. Rebuild or replace the projection workflow from `model_projection_inputs/source_files` so heterogeneous permeability fields are mesh-cell data with documented field names and reproducible generation.
13. Prepare one OGS run directory per permeability-field sample with `inversion_workflow/scripts/prepare_ogs_run.py`; each run must carry a `RUN_MANIFEST.json` recording the generated or fitted field and the output variables required by observation operators.
14. Execute the prepared run with `inversion_workflow/scripts/run_ogs_model.py` using either a host OGS executable, a host Apptainer/Singularity runtime, or the recorded Dockerized-Apptainer path for the collected OGS 6.5.4 SIF, and keep `OGS_EXECUTION_STATUS.json` with the run outputs.

15. Sample OGS outputs with `sample_ogs_state_outputs.py`, evaluate the state target layer with `evaluate_state_observation_targets.py`, and assemble the direct permeability and state terms with `assemble_inversion_objective.py`.
16. Use `evaluate_inversion_candidate.py` as the per-candidate harness. The verified runs on the current direct-fit and smooth-search meshes prepare run copies, audit run-input mesh fields and submeshes, evaluate direct permeability, execute or record the OGS command, sample available state outputs, audit the RH boundary curve, evaluate state targets, and write the combined objective summary.
17. Use `run_inversion_candidate_search.py` as the deterministic search loop around that harness. The current dry search ranks four smooth candidates and keeps `length_0p025m` first with objective 269.97. In dry mode this is still a direct-permeability-only ranking; in execute mode with the collected SIF through Dockerized Apptainer, the same loop ranks by the combined permeability plus state-observation objective and can be replaced by a Bayesian optimizer or ensemble sampler.
18. Use `rank_regularized_permeability_candidates.py` to choose a small first OGS comparison set from the direct-only candidates. At minimum this should include the direct-misfit winner and one damped regularized candidate, because the state observations are needed to decide whether the strongest permeability update is physically acceptable.
19. Prepare that comparison set with `run_regularized_ogs_candidate_set.py`. The current executed set contains the full-shift, 0.75-shift, and 0.5-shift 0.025 m fields, each with audited run inputs, 83 OGS output VTUs, sampled NMR rows, and a combined objective summary. The first adaptive broader-support batch has also been executed and is worse than the then-best 0.025 m field. The first local-refinement batch improves the smooth-field best to `length_0p013m` with combined objective 3159.66. The first local-bracketing batch around that field is worse than this best, with best combined objective 3167.21. The finite-candidate Bayesian proposal batch is also worse, with best combined objective 3172.37. The first continuous proposal batch improves the active best to `length_0p006m_shift_0p992` with combined objective 3156.78. The lower-support continuous batch further improves the active best to `length_0p004m_shift_0p995` with combined objective 3156.38, and the first repeatable lower-support loop improves it again to `length_0p003m_shift_1p006` with combined objective 3156.37. The second repeatable lower-support loop did not improve this incumbent, and the refreshed broad continuous evidence batches also did not improve it; the best broad-loop row is `length_0p010m` with combined objective 3157.70. The next step is to treat `basis_024_det_1_0p0075_s_1p000` as the current incumbent evidence. Five production handoff rounds did not improve it; the local-anisotropy screen and cross-stream hybrid screen also found no direct-improving follow-up family. Use `PRODUCTION_SAMPLER_DECISION.md` as the current decision to pause the active direct-permeability plus NMR production sampler and move effort to gated measurement streams or a new field family.
20. Reproduce the current open-niche pressure curve from RH/T data, or document the residual mismatch if the source data do not fully reproduce it.
21. Start with a heterogeneous anisotropic permeability field and fixed porosity, retention, elasticity, swelling, and thermal properties.
22. Use pulse-test data as direct log-permeability constraints with large interval-scale errors and gas/slip caveats.
23. Use ERT through the documented theta-to-resistivity operator after spatial projection to the ERT/common mesh exists; use NMR as a sparse water-content anchor with bound-water bias/error.
24. Use Taupe/TDR through the documented trend operator and suction/RH as trend/consistency constraints until their calibration to model saturation is documented.
25. Only after permeability is identifiable should porosity, m , p_b , or other fields be released as additional unknowns.

10 Full XML listings

04_media_TRM.xml

```
<!--
Parameter, die in Ogs5 vorhanden waren, welche in Ogs6 bisher aber noch zu fehlen scheinen
oder
nicht mehr benötigt werden:
PERMEABILITY SATURATION
CAPILLARY PRESSURE
STORAGE
-->
<medium id="0,1"><!-- ,2,3,4,6,7,8,9,11,12,13,14"-->
  <properties>
    <property>
```

```

        <name>permeability</name>
        <type>Parameter</type>
        <parameter_name>k_i</parameter_name>
    </property>
</property>
    <name>porosity</name>
    <type>Parameter</type>
    <parameter_name>phi</parameter_name>
</property>
<property>
    <name>thermal_conductivity</name>
    <type>EffectiveThermalConductivityPorosityMixing</type>
</property>
<property>
    <name>biot_coefficient</name>
    <type>Parameter</type>
    <parameter_name>biot</parameter_name>
</property>
    <!-- zusätzliche TRM Parameter -->
    <include file="./04_2_media_twophase.xml" />
</properties>
<phases>
    <!--phase:
    AqueousLiquid-->
    <include file="./04_1_media_aqu_liq.xml" />
    <!--phase:
    AqueousLiquid-->
    <phase>
        <type>Solid</type>
        <properties>
            <property>
                <name>density</name>
                <type>Parameter</type>
                <parameter_name>rho_s</parameter_name>
            </property>
            <property>
                <name>thermal_conductivity</name>
                <type>Parameter</type>
                <parameter_name>kappa_th_s</parameter_name>
            </property>
            <property>
                <name>specific_heat_capacity</name>
                <type>Parameter</type>
                <parameter_name>c_p_s</parameter_name>
            </property>
            <property>
                <name>thermal_expansivity</name>
                <type>Parameter</type>
                <parameter_name>CTE</parameter_name>
            </property>
            <property>
                <name>swelling_stress_rate</name>
                <type>SaturationDependentSwelling</type>
                <swelling_pressures>1.0E+06 1.0E+06 1.0E+06</swelling_pressures>
                <exponents>1 1 1</exponents>
                <lower_saturation_limit>0.1</lower_saturation_limit>
                <upper_saturation_limit>1</upper_saturation_limit>
            </property>
        </properties>
    </phase>
</phases>
</medium>

```

04_1_media_aqu_liq.xml

```
<phase>
```

```

<type>AqueousLiquid</type>
<properties>
  <property>
    <name>specific_heat_capacity</name>
    <type>Parameter</type>
    <parameter_name>c_p_1</parameter_name>
  </property>
  <property>
    <name>thermal_conductivity</name>
    <type>Parameter</type>
    <parameter_name>kappa_1</parameter_name>
  </property>
  <property>
    <name>viscosity</name>
    <type>Parameter</type>
    <parameter_name>eta</parameter_name>
  </property>
  <property>
    <name>density</name>
    <type>Linear</type>
    <reference_value>1095</reference_value>
    <independent_variable>
      <variable_name>temperature</variable_name>
      <reference_condition>298.15</reference_condition>
      <slope>-5E-4</slope>
    </independent_variable>
    <independent_variable>
      <variable_name>liquid_phase_pressure</variable_name>
      <reference_condition>7.5e6</reference_condition>
      <slope>3.2e-10</slope>
    </independent_variable>
  </property>
  <!--<property>
    <name>density</name>
    <type>Function</type>
    <value>
      <expression>730.6 + 2.025*temperature -3.8e-3*temperature*temperature + (2.362-1.197
        e-2*temperature+1.835e-5*temperature*temperature)*phase_pressure*1e
        -6+(2.374-1.024e-2*temperature+1.49e-5*temperature*temperature)*150</expression
      >
    </value>
    <dvalue>
      <variable_name>temperature</variable_name>
      <expression>2.025-7.6e-3*temperature +(-1.197e-2+3.67e-5*temperature)*
        phase_pressure*1e-6 +(-1.024e-2+2.98e-5*temperature)*150</expression>
    </dvalue>
    <dvalue>
      <variable_name>phase_pressure</variable_name>
      <expression>(2.362-1.197e-2*temperature+1.835e-5*temperature*temperature)*1e-6</
        expression>
    </dvalue>
  </property>-->
  <!-- <property>
    <name>viscosity</name>
    <type>Function</type><value>
      <expression>1e-3*((1+2.765e-3*150)*exp(11.897-5.943e-2*temperature+6.422e-5*
        temperature*temperature))</expression>
    </value>
    <dvalue><variable_name>temperature</variable_name>
      <expression>exp((-0.05943 + 0.00006422*temperature)* temperature)*(-12.3448 +
        0.0266795*temperature)</expression>
    </dvalue>
  </property> -->
  <!-- <property>
    <name>vapour_density</name>
    <type>WaterVapourDensity</type>
  </property> -->
  <!-- <property>
    <name>vapour_diffusion</name>
    <type>VapourDiffusionFEBEX</type>

```

```

        <tortuosity>1.0</tortuosity>
    </property> -->
    <!--<property>
        <name>latent_heat</name>
        <type>LinearWaterVapourLatentHeat</type>
    </property>-->
    <!--
        <property>
            <name>thermal_diffusion_enhancement_factor
            </name>
            <type>Constant</type>
            <value>1.0</value>
        </property> -->
    </properties>
</phase>

```

04_2_media.xml

```

<property>
    <name>relative_permeability</name>
    <type>RelativePermeabilityVanGenuchten</type>
    <residual_liquid_saturation>0.1</residual_liquid_saturation>
    <residual_gas_saturation>0</residual_gas_saturation>
    <exponent>0.45</exponent>
    <minimum_relative_permeability_liquid>1e-25</minimum_relative_permeability_liquid>
</property>
<property>
    <name>saturation</name>
    <type>SaturationVanGenuchten</type>
    <residual_liquid_saturation>0.1</residual_liquid_saturation>
    <residual_gas_saturation>0</residual_gas_saturation>
    <exponent>0.45</exponent>
    <p_b>10e6</p_b>
</property>
<property>
    <name>bishops_effective_stress</name>
    <!--type>BishopsSaturationCutoff</type>
    <cutoff_value>1</cutoff_value-->
    <type>BishopsPowerLaw</type>
    <exponent>1</exponent>
</property>

```

03_parameters_TRM.xml

```

<!-- IC -->
<parameter>
    <name>ic_displacement</name>
    <type>Constant</type>
    <values>0 0</values>
    <!--type>MeshNode</type>
    <field_name>displacement</field_name-->
</parameter>
<parameter>
    <name>ic_pressure</name> <!-- check pressure0 + ic_sigma0 -->
    <type>Function</type>
    <expression>1.5e6 - ((1095*(1 + 3.2e-10*(1.5e6 - 7.5e6))))*0*y</expression> <!-- just
        1% difference if not considering the change in rho-->
    <!--type>MeshNode</type>
    <field_name>pressure_interpolated</field_name-->
</parameter>
<parameter>
    <name>ic_temperature</name> <!-- 25°C -->
    <type>Constant</type>
    <value>298.15</value>
    <!--type>MeshNode</type>
    <field_name>temperature_interpolated</field_name-->
</parameter>

```

```

<parameter>
  <name>ic_sigma0</name><!-- check in-situ pressure -->
  <type>Function</type> <!-- values from discussion between BGR and GRS -->
  <expression>(-1.21e6) + 1*((1095*(1 + 3.2e-10*(1.5e6 - 7.5e6))))*0*y</expression> <!--
    \nu_13 0.22 * \sigma_v'-->
  <!--expression>(-2.42e6) + 1*((1095*(1 + 3.2e-10*(1.5e6 - 7.5e6))))*0*z</expression-->
  <!-- \nu_23 0.44 * \sigma_v'-->
  <expression>(-7e6 + 1*1.5e6) + 1*((1095*(1 + 3.2e-10*(1.5e6 - 7.5e6))))*0*y</
    expression> <!-- sigma_v' -->
  <expression>0</expression>
  <expression>0</expression>
  <!--expression>0</expression-->
</parameter>
<!-- BC -->
<parameter>
  <name>bc_displacement</name>
  <type>Constant</type>
  <value>0</value>
</parameter>
<parameter>
  <name>bc_pressure_outside</name> <!-- check imposed ic_press* -->
  <type>Constant</type>
  <value>1.5e+6</value>
</parameter>
<parameter>
  <name>bc_top_pressure</name>
  <type>Function</type>
  <expression>1.5e6 - ((1095*(1 + 3.2e-10*(1.5e6 - 7.5e6))))*0*50</expression>
</parameter>
<parameter>
  <name>bc_temperature_outside</name> <!-- check value ic_temp* -->
  <type>Constant</type>
  <value>298.15</value>
</parameter>
<parameter>
  <name>load_top</name>
  <type>Function</type>
  <expression>(-7e6) + 1*((1095*(1 + 3.2e-10*(1.5e6 - 7.5e6))))*0*50</expression> <!--
    use total stress for consistency here; load_top at z=30m -->
</parameter>
<!-- Boundary conditions for gallery and niche -->
<parameter>
  <name>bc_atm_pressure</name>
  <type>Constant</type>
  <values>0.1e+6</values>
</parameter>
<parameter>
  <name>pressure_scaling_factor</name>
  <type>Constant</type>
  <value>1.0</value>
</parameter>

<parameter>
  <name>open_niche_seasonal</name>
  <type>CurveScaled</type>
  <curve>open_niche_seasonal_curve</curve>
  <parameter>pressure_scaling_factor</parameter>
</parameter>

<!--
<parameter>
  <name>closed_niche_seasonal</name>
  <type>CurveScaled</type>
  <curve>closed_niche_seasonal_curve</curve>
  <parameter>pressure_scaling_factor</parameter>
</parameter>
-->

<!--parameter phase solid -->
<parameter>

```



```

    <name>rho_s</name> <!--dry
    solid density [kg.m3]-->
    <type>Constant</type>
    <values>2.52e+3</values>
</parameter>
<parameter> <!-- check -->
    <name>kappa_th_s</name> <!--thermal
    conductivtiy [W/mK]-->
    <type>Constant</type>
    <values>2.156</values>
</parameter>
<parameter> <!-- check -->
    <name>c_p_s</name> <!--specific
    head capacity [J/kgK]-->
    <type>Constant</type>
    <values>1254.74</values>
</parameter>
<parameter> <!-- check -->
    <name>CTE</name> <!-- coefficient of linear thermal_expansivity [J/kgK]-->
    <type>Constant</type>
    <values>1254.74</values>
</parameter>
<!-- parameter phase liquid -->
<parameter> <!-- check -->
    <name>c_p_l</name> <!--specific
    head capacity liquid [J/kgK]-->
    <type>Constant</type>
    <values>4160</values>
</parameter>
<parameter> <!-- check -->
    <name>kappa_l</name> <!--thermal
    conductivity [W/mK]-->
    <type>Constant</type>
    <values>0.6</values>
</parameter>
<parameter> <!-- check -->
    <name>eta</name> <!--viscosity
    [J/kgK]-->
    <type>Constant</type>
    <value>1.0e-3</value>
</parameter>
<!-- parameter medium -->
<parameter>
    <name>phi</name> <!--porosity
    [-] check-->
    <type>Constant</type>
    <values>0.105</values>
</parameter>
<parameter>
    <name>biot</name> <!--Biot
    parameter [-]-->
    <type>Constant</type>
    <values>1</values>
</parameter>
<parameter> <!-- check -->
    <name>k_i</name> <!--intrinsic
    permeability [m2]-->
    <type>Constant</type>
    <!--values>1.00E-19 0.00 0.00 0.00 0.40E-19 0.00 0.00 0.00 1.00E-19</values-->
    <values>1.00E-19 0.00 0.00 0.40E-19</values>
    <!-- <values>2.6666666666666663e-20</values> -->
</parameter>
<parameter>
    <name>E</name>
    <type>Constant</type>
    <values>13.80e+9 6.00e+9 13.80e+9</values>
    <!-- <values>11.20e+9</values> -->
</parameter>
<parameter>
    <name>G</name>

```

```

    <type>Constant</type>
    <values>4.79e+9 2.46e+9 4.79e+9</values>
    <!-- <values>4.013e+9</values> -->
</parameter>
<parameter>
    <name>nu</name>
    <type>Constant</type>
    <values>0.44 0.22 0.44</values>
    <!-- <values>0.37</values> -->
</parameter>
<!--parameter>
    <name>e0</name>
    <type>Constant</type>
    <values>1.0000E+00 0.0000E+00 0.0000E+00</values>
</parameter>
<parameter>
    <name>e1</name>
    <type>Constant</type>
    <values>0.0000E+00 8.3867E-01 -5.4464E-01</values>
</parameter>
<parameter>
    <name>e2</name>
    <type>Constant</type>
    <values>0.0000E+00 5.4464E-01 8.3867E-01</values>
</parameter-->

```

References

- [1] Florian Amann, Katja M. Wild, Simon Loew, Salina Yong, Reto Thöny, and Edwin Frank, *Geomechanical behaviour of opalinus clay at multiple scales: results from mont terri rock laboratory (switzerland)*, Swiss Journal of Geosciences **110** (2017), no. 1, 151–171, Fulltext saved in the project Library/fulltexts folder.
- [2] G. E. Archie, *The electrical resistivity log as an aid in determining some reservoir characteristics*, Transactions of the AIME **146** (1942), no. 1, 54–62, Fulltext saved in the project Library/fulltexts folder.
- [3] Shuang Chen, Gesa Ziefle, and Tuanny Cajuhi, *BGR modelling activities under the EURAD and CD-A projects*, Local BGR technical-discussion slide deck received through the CD-A email/Thunderbird data collection, 2026, Local fulltext copied to the project Library/fulltexts folder; exact source path recorded in the Library source coverage audit.
- [4] Stephan Costabel, *Single-sided nuclear magnetic resonance on the walls in the CD-A niches*, Local BGR slide deck received through the CD-A email/Thunderbird data collection, 2026, Local fulltext copied to the project Library/fulltexts folder; exact source path recorded in the Library source coverage audit.
- [5] Yingzhi Cui, Igor Shikhov, and Christoph H. Arns, *Nmr relaxation modelling in porous media with dual-scale-resolved internal magnetic fields*, Transport in Porous Media **142** (2022), 453–474, Open-access Springer fulltext saved in the project Library/fulltexts folder.
- [6] Benoît Garitte, Trung Son Nguyen, Jean-Denis Barnichon, Björn Graupner, Chae Lee, Koji Maekawa, et al., *Modelling the mont terri he-d experiment for the thermo-hydraulic-mechanical response of a bedded argillaceous formation to heating*, Environmental Earth Sciences **76** (2017), no. 9, 345, Fulltext saved in the project Library/fulltexts folder.
- [7] Andreas Gautschi, *Safety-relevant hydrogeological properties of the claystone barrier of a swiss radioactive waste repository: An evaluation using multiple lines of evidence*, Grundwasser **22** (2017), no. 3, 221–233, Fulltext saved in the project Library/fulltexts folder.
- [8] Antonio Gens, Jean Vaunat, Benoît Garitte, and Yannick Wileveau, *In situ behaviour of a stiff layered clay subject to thermal loading: observations and interpretation*, Géotechnique **57** (2007), no. 2, 207–228, Fulltext saved in the project Library/fulltexts folder from the UPC open-access repository.
- [9] Antonio Gens, Klaus Wieczorek, Irina Gaus, Benoît Garitte, Juan Carlos Mayor, Kristof Schuster, Gilles Armand, José Luis García-Siñeriz, and Thomas Trick, *Performance of the opalinus clay under thermal loading: experimental results from mont terri rock laboratory (switzerland)*, Swiss Journal of Geosciences **110** (2017), 269–286, Fulltext saved in the project Library/fulltexts folder.
- [10] R. L. Kleinberg, *Utility of nmr t2 distributions, connection with capillary pressure, clay effect, and determination of the surface relaxivity parameter rho2*, Magnetic Resonance Imaging **14** (1996), no. 7–8, 761–767, Fulltext not downloaded; see the project Library unavailable-fulltexts log.

- [11] L. J. Klinkenberg, *The permeability of porous media to liquids and gases*, Drilling and Production Practice (1941), 200–213, Fulltext saved in the project Library/fulltexts folder.
- [12] Franz Königer, *CD-A: Differential TDR using TAUPE sensor cables*, Local ISU/BGR slide deck received through the CD-A email/Thunderbird data collection, 2026, Local fulltext copied to the project Library/fulltexts folder; exact source path recorded in the Library source coverage audit.
- [13] Peter Marschall, Stanley Horseman, and Thomas Gimmi, *Characterisation of gas transport properties of the opalinus clay, a potential host rock formation for radioactive waste disposal*, Oil & Gas Science and Technology – Revue de l’IFP **60** (2005), no. 1, 121–139, Fulltext saved in the project Library/fulltexts folder.
- [14] OpenGeoSys Community, *Parameters*, <https://www.opengeosys.org/stable/docs/userguide/blocks/parameters/>, 2025, Official OGS user guide, section “MeshElement and MeshNode”, accessed 2026-05-28.
- [15] ———, *Thermo-richards-mechanics process*, <https://www.opengeosys.org/docs/processes/thermal-processes/trm/>, 2025, Official OGS documentation, section “Governing equations without vapor diffusion”, accessed 2026-05-28.
- [16] A. Revil, L. M. Cathles, S. Losh, and J. A. Nunn, *Electrical conductivity in shaly sands with geophysical applications*, Journal of Geophysical Research: Solid Earth **103** (1998), no. B10, 23925–23936, Cornell-hosted fulltext saved in the project Library/fulltexts folder.
- [17] D. A. Robinson, S. B. Jones, J. M. Wraith, D. Or, and S. P. Friedman, *A review of advances in dielectric and electrical conductivity measurement in soils using time domain reflectometry*, Vadose Zone Journal **2** (2003), no. 4, 444–475, USDA-hosted fulltext saved in the project Library/fulltexts folder.
- [18] Senecio Schefer and Sebastian Condamin, *Precision levelling twin niches*, Local swisstopo technical-discussion slide deck received through the CD-A email/Thunderbird data collection, 2026, Local fulltext copied to the project Library/fulltexts folder; exact source path recorded in the Library source coverage audit.
- [19] Louise Theodon and BGR CD-A contributors, *Input HERMES BGR: CD-A measurement-driven modelling input note*, Local BGR/HERMES note received through the CD-A email data collection, 2024, Local fulltext copied to the project Library/fulltexts folder; exact source path recorded in the Library source coverage audit.
- [20] William Thomson, *On the equilibrium of vapour at a curved surface of liquid*, Philosophical Magazine **42** (1871), no. 282, 448–452, Primary Kelvin-equation source; fulltext saved in the project Library/fulltexts folder from Zenodo record 1742066.
- [21] G. C. Topp, J. L. Davis, and A. P. Annan, *Electromagnetic determination of soil water content: Measurements in coaxial transmission lines*, Water Resources Research **16** (1980), no. 3, 574–582, Fulltext not downloaded; see the project Library/unavailable-fulltexts log.
- [22] M. Th. van Genuchten, *A closed-form equation for predicting the hydraulic conductivity of unsaturated soils*, Soil Science Society of America Journal **44** (1980), no. 5, 892–898, Fulltext saved in the project Library/fulltexts folder.
- [23] Luc R. Van Loon and Josep M. Soler, *Diffusion of hto, $^{36}\text{cl}^-$, $^{125}\text{i}^-$ and $^{22}\text{na}^+$ in opalinus clay: Effect of confining pressure, sample orientation, sample depth and temperature*, Tech. Report Nagra Technical Report NTB 03-07, National Cooperative for the Disposal of Radioactive Waste, Wettingen, Switzerland, 2003, Fulltext saved in the project Library/fulltexts folder.
- [24] María Victoria Villar and Fernando Javier Romero, *Water retention curves of opalinus clay*, Tech. Report CIEMAT Informe Técnico 1262, CIEMAT, Madrid, Spain, 2012, Fulltext saved in the project Library/fulltexts folder.
- [25] Wenqing Wang, Georg Kosakowski, and Olaf Kolditz, *A parallel finite element scheme for thermo-hydro-mechanical (thm) coupled problems in porous media*, Computers & Geosciences **35** (2009), no. 8, 1631–1641, Fulltext saved in the project Library/fulltexts folder.
- [26] Gesa Zieffle, *CD-A for HERMES 2d: modelling discussion slides*, Local BGR/Gesa slide deck received through the CD-A email/TeamBeam data collection, 2025, Local fulltext copied to the project Library/fulltexts folder; exact source path recorded in the Library source coverage audit.
- [27] ———, *TD517 CD-A: technical discussion minutes from 2026-05-07*, Local meeting minutes received through the CD-A email/Thunderbird data collection, 2026, Local fulltext copied to the project Library/fulltexts folder; exact source path recorded in the Library source coverage audit.
- [28] Gesa Zieffle, Tuanny Cajuhi, Stephan Costabel, Antoine Fourrière, Markus Furche, Jana Gerowski, Tilo Kneuker, Franz Königer, and Jobst Maßmann, *Cd-a twin niches in the mont terri rock laboratory: Characterization and interpretation of hydraulic parameters with regard to safety aspects*, International Journal of Rock Mechanics and Mining Sciences **174** (2024), 105624, Fulltext saved in the project Library/fulltexts folder.

- [29] Gesa Ziefle, Tuanny Cajuhi, Stephan Costabel, Markus Furche, and Jobst Maßmann, *Water content evolution in the edz of opalinus clay: A methodic approach for a comparative interpretation of measurements and modelling*, Rock Mechanics and Rock Engineering **57** (2024), 4443–4472, Fulltext saved in the project Library/fulltexts folder.